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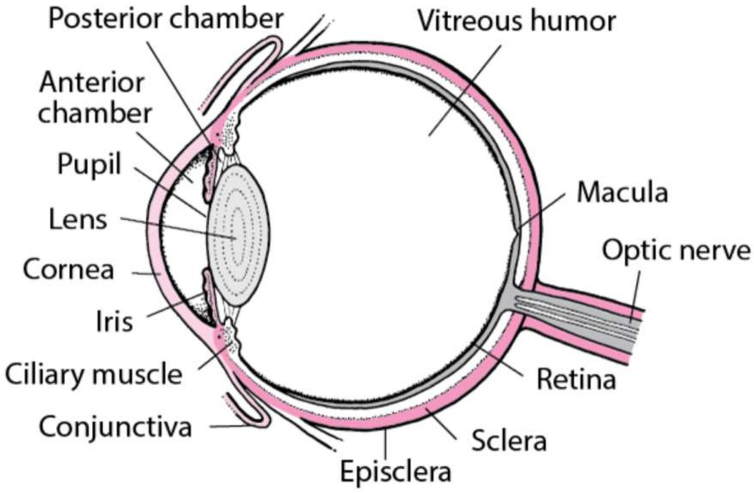
HY Ophthal, by Dr Michael D Mehlman

The aim of this PDF is not to teach you every ophthal anatomical point and pathologic condition. The purpose is to teach you exactly what you need to know for USMLE – that's it. This PDF is for all Steps 1, 2CK, and 3.

First I present HY info in chart/table format. Then there are 50 questions to test your understanding.

I cover some concepts in the 50 questions that are not covered in the charts/tables, so I recommend doing both.

Basic anatomy of eye:



Anterior chamber: Fluid-filled space between the cornea and iris; contains aqueous humor, a clear fluid that nourishes the cornea and maintains intraocular pressure.

Aqueous humor: Clear, watery fluid in the anterior and posterior chambers; serves to maintain the shape of the eye, deliver nutrients to the avascular cornea and lens, and help regulate intraocular pressure.

Ciliary muscle: Ring-shaped smooth muscle located just posterior to the iris; contraction and relaxation alter the shape of the lens, allowing it to focus on objects at varying distances (i.e., accommodation).

Conjunctiva: Thin, transparent mucous membrane covering the anterior surface of the eye (i.e., bulbar conjunctiva) and lines the inner surface of the eyelids (i.e., tarsal / palpebral conjunctiva); serves to protect and lubricate the eye, producing mucus and tears to maintain moisture and prevent foreign particles from entering the eye.

Cornea: Transparent anterior surface of the eye that covers the iris and pupil; plays a crucial role in focusing incoming light onto the retina, allowing for visual acuity.

Iris: Circular structure posterior to the cornea that controls size of the pupil; pigmentation gives it color.

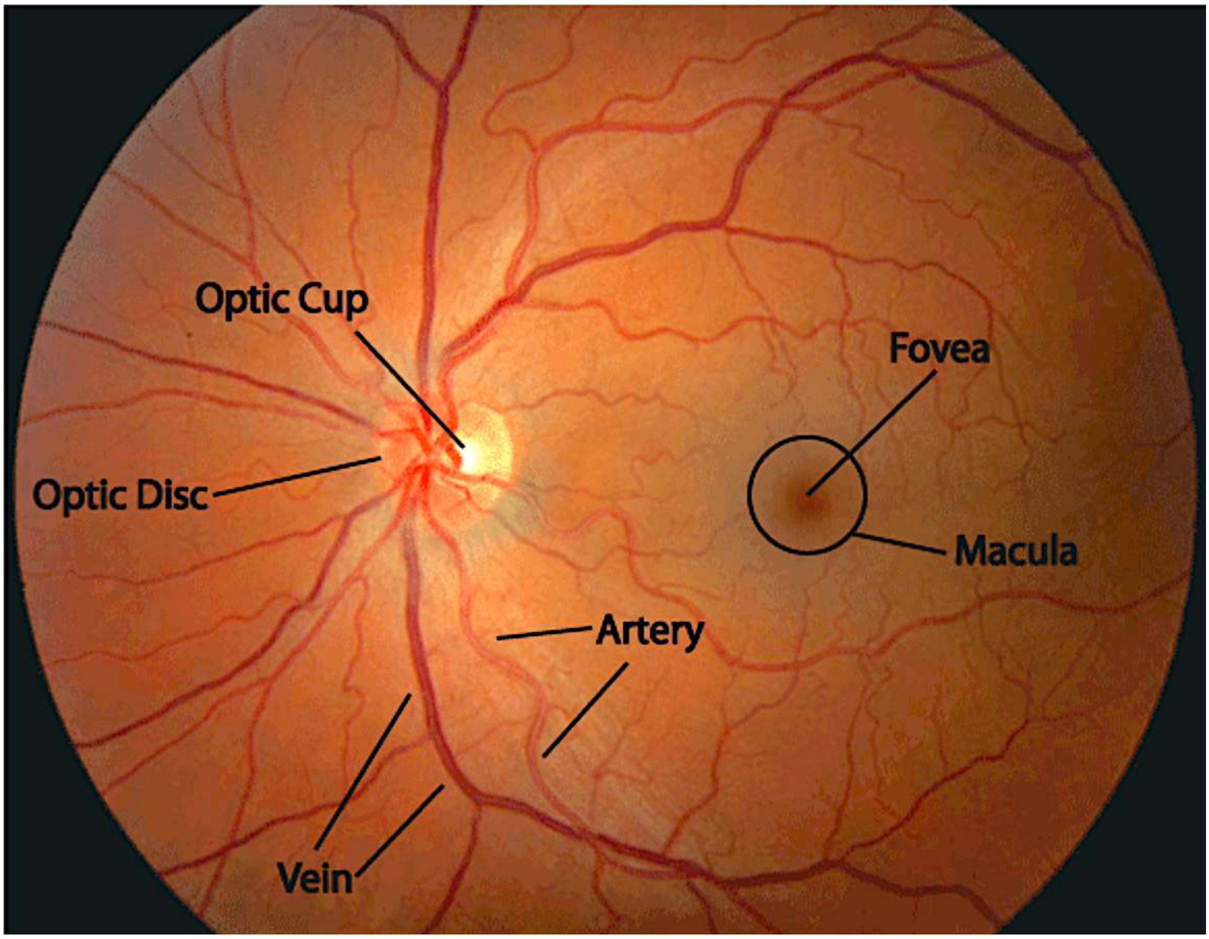
Posterior chamber: Fluid-filled between the iris and front of the lens; contains watery fluid called aqueous humor, which helps maintain intraocular pressure and nourishes the cornea and lens.

Pupil: Circular, black opening at the center of the iris that regulates the amount of light entering the eye.

Sclera: Tough outermost layer of the eye that forms the visible white of the eye; provides structural support and protection to the internal structures of the eye; contributes to the eye's shape and integrity. The episclera is a thin, transparent layer just deep to the conjunctiva that contains blood vessels and helps nourish the sclera below.

Vitreous humor: Transparent, gel-like substance that fills the posterior segment of the eye between the lens and retina; helps maintain the shape of the eye, assists in transmitting light to the retina, and provides structural support to the retina.

Fundoscopic hyper-basics for USMLE:



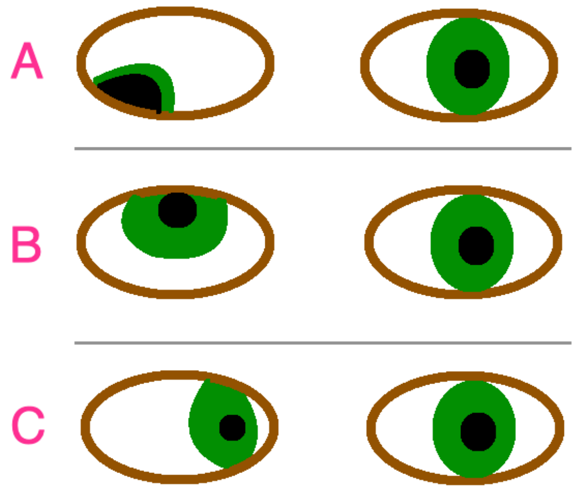
Optic disc: Structure at the back of the eye that serves as the exit point for the optic nerve.

Optic cup: Depression within optic disc where the retinal ganglion cells converge to form the optic nerve.

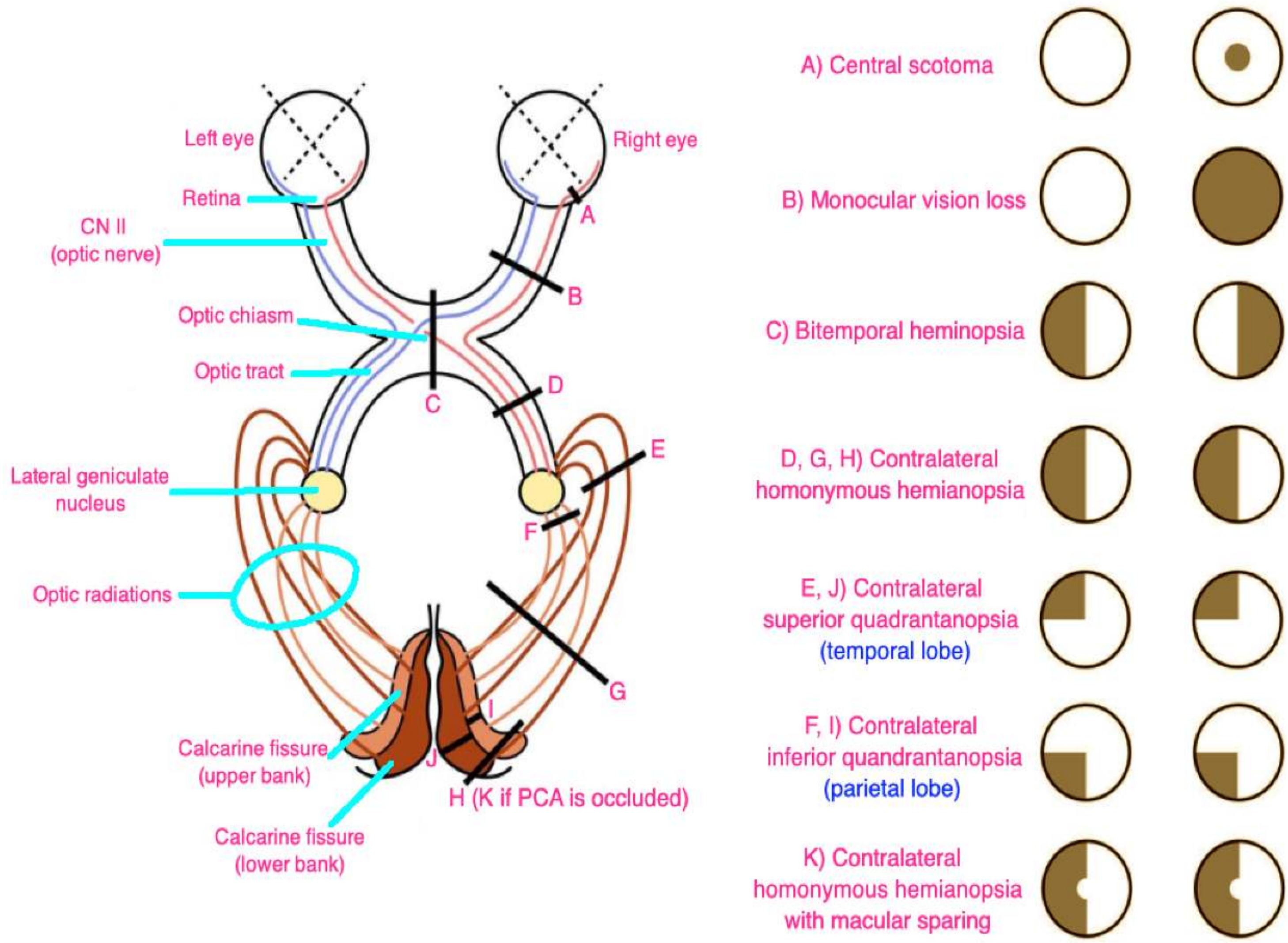
Macula: Central depression at back of retina with increased numbers of cone cells, which are required for color vision and visual acuity.

Fovea: Central pit within the macula with an even greater density of cone cells; enables sharp, central vision.

Extraocular nerve palsies



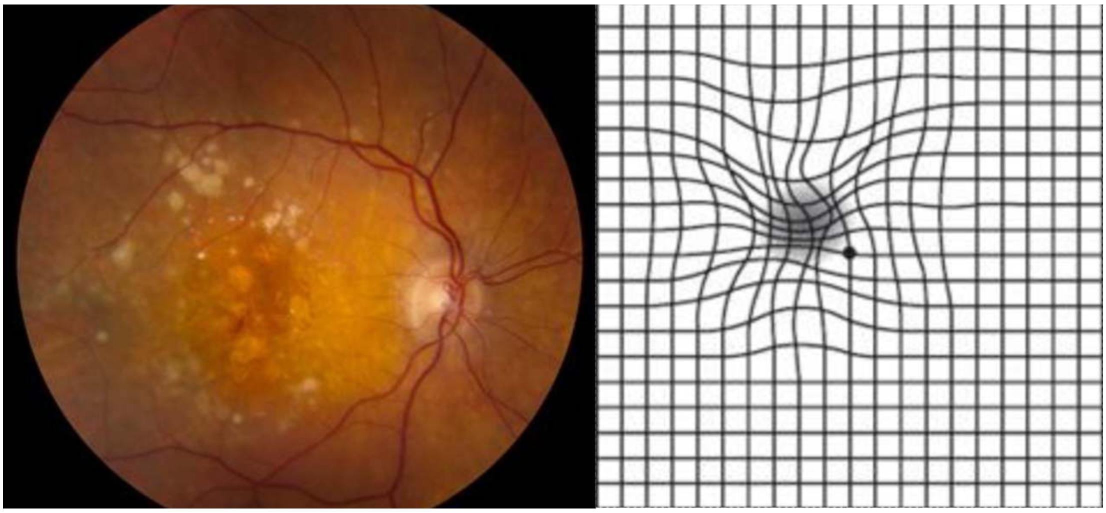
- A = CN III (oculomotor) lesion → presents as down and out.
- B = CN IV (trochlear) lesion → presents as upward gaze + patient will tilt head toward unaffected side.
- C = CN VI (abducens) lesion → medially directed eye.



Glaucoma	
Open-angle	- Chronic eye condition characterized by increased intraocular pressure due to impaired drainage of aqueous humor. - Unlike acute angle-closure (closed-angle) glaucoma, the drainage angle (canal of Schlemm) between the cornea and iris remains open. - Progresses slowly and is usually painless. - As it advances, causes peripheral vision loss (tunnel vision) and results in optic nerve damage characterized by “cupping of the optic disc” (i.e., an increased cup-to-disc ratio). - The USMLE wants **tonometry** as the first step in management (measures intraocular pressure). - Treatment for open-angle glaucoma for USMLE is not going to be a trivia game of “which drug first.” They just want you to know that topical prostaglandins, such as **latanoprost,** can increase outflow of aqueous humor, thereby reducing IOP. - Topical beta-blockers, such as **timolol,** decrease production of aqueous humor. **- Carbonic anhydrase inhibitors,** such as dorzolamide, decrease production of aqueous humor. - **Pilocarpine** is a muscarinic receptor agonist that constricts the pupil and increases outflow of aqueous humor.
Closed-angle	- Presents on USMLE as a very buzzy red, teary, painful eye with a fixed mid-dilated pupil. - Due to closure of the drainage angle (canal of Schlemm) between the cornea and the iris, leading to a rapid increase in IOP. - Can lead to compression of the optic nerve and blood vessels, causing severe symptoms such as intense eye pain, headache, nausea, vomiting, blurred vision, and seeing colored rings or halos. - Similar to open-angle glaucoma, the USMLE wants **tonometry** as the first step to confirm increased IOP. - Treatment is IV mannitol (helps draw excess aqueous humor out of the eye), timolol, pilocarpine (muscarinic receptor agonist), or alpha-2 agonists such as brimonidine.

Macular degeneration

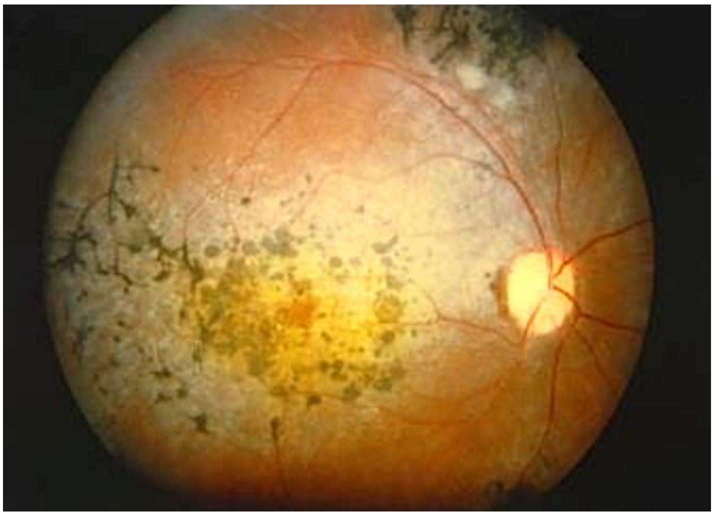
- USMLE wants you to know that “wavy lines” on a straight-line Amsler test = macular degeneration.



- Characterized by loss of central vision.
- Fundoscopy shows drusen, which are yellow lipoproteinaceous deposits.
- There are dry (atrophic) and wet (neovascular) types of macular degeneration.
- Most commonly occurs idiopathically due to age (i.e., age-related macular degeneration).
- Dry is treated supportively; wet is treated with anti-VEGF agents and sometimes laser therapy.

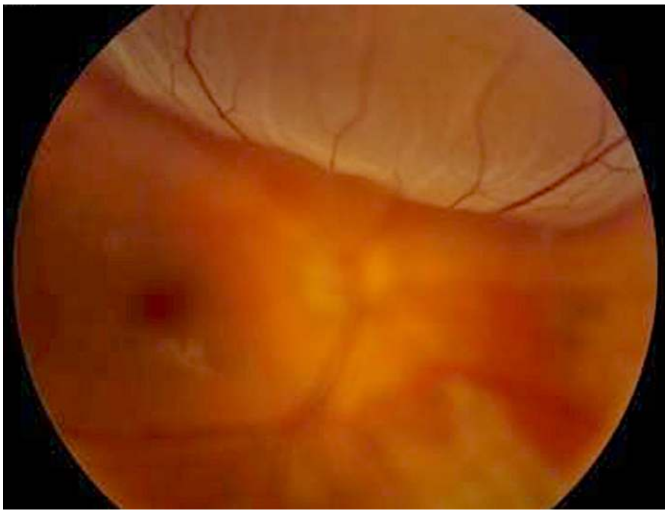
Retinitis pigmentosa

- Presents on USMLE as a combo of nyctalopia (loss of night vision) and loss of peripheral vision.
- There will often be a family history. Fundoscopy shows characteristic pigmentation.



Retinal detachment

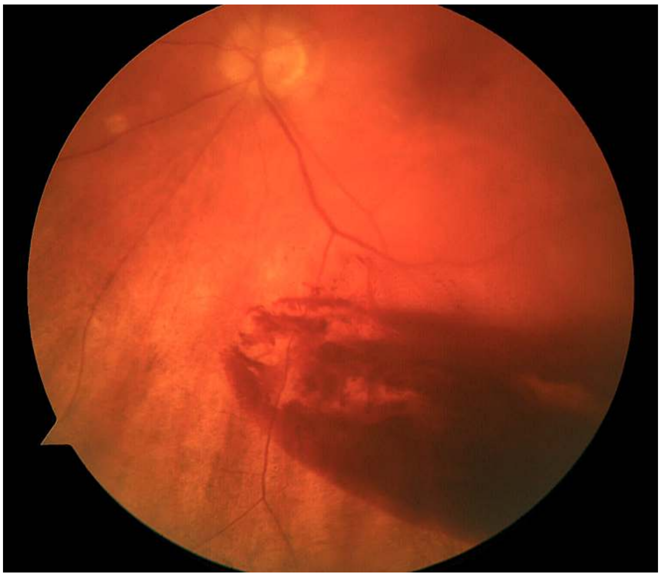
- Occurs classically in those with sudden trauma to the eye or severe myopia.
- The fundoscopy is HY and important for USMLE.



- Treatment is surgical and includes pneumatic retinopexy (i.e., gas bubble is injected into the eye to push the retina back into place; scleral buckling (piece of silicone is attached to the sclera to exert posteriorly-directed pressure so the retina reattaches); and vitrectomy (vitreous humor is removed and replaced with gas or silicone oil to flatten the retina).

Vitreous hemorrhage

- USMLE wants you to know that the biggest risk factors for bleeding into the vitreous are diabetes and hypertension.

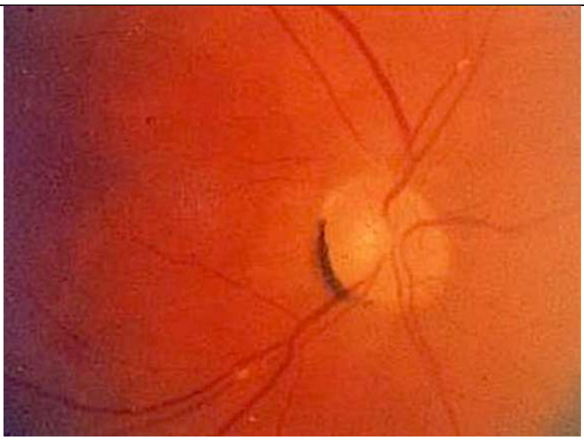


- Vitreous hemorrhage will usually be a distractor, where they can tell you a young boxer was hit in the eye, but the answer is retinal detachment, not vitreous hemorrhage.

Central retinal artery occlusion

- Presents as sudden, painless vision loss in a patient where a "curtain comes down."
- Due to 1) carotid atheroma (in setting of HTN) that's launched off to the eye, or 2) left atrial mural thrombus (in setting of atrial fibrillation) that's launched off to the eye.
- A pale retina will be seen on fundoscopy with a cherry red spot due to maintenance of perfusion from the choroidal artery.

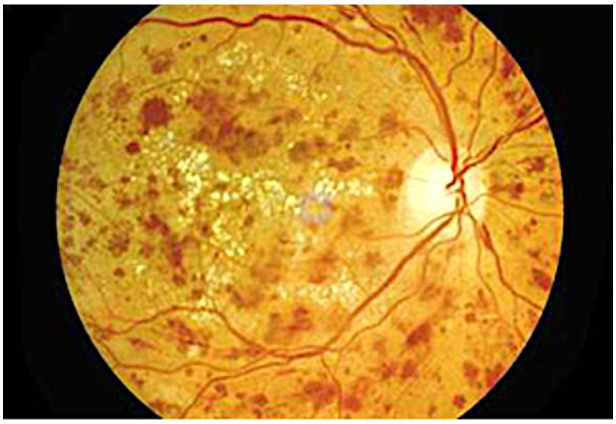
- Ocular massage can be attempted as Tx, where the embolus may be dislodged, restoring perfusion.

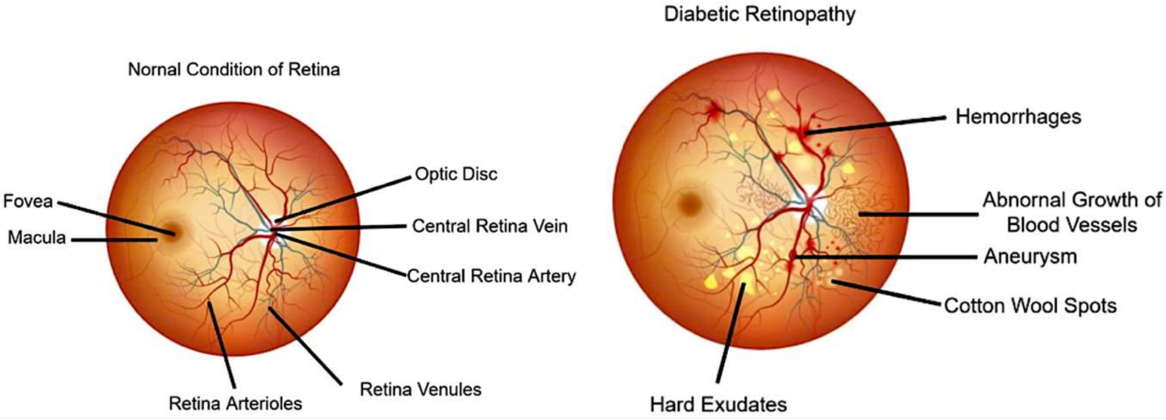


- USMLE can show image that looks similar to this one. They want you to know that weird crescent-shaped finding is called a "refractile body," and refers to an embolus.

Diabetic retinopathy

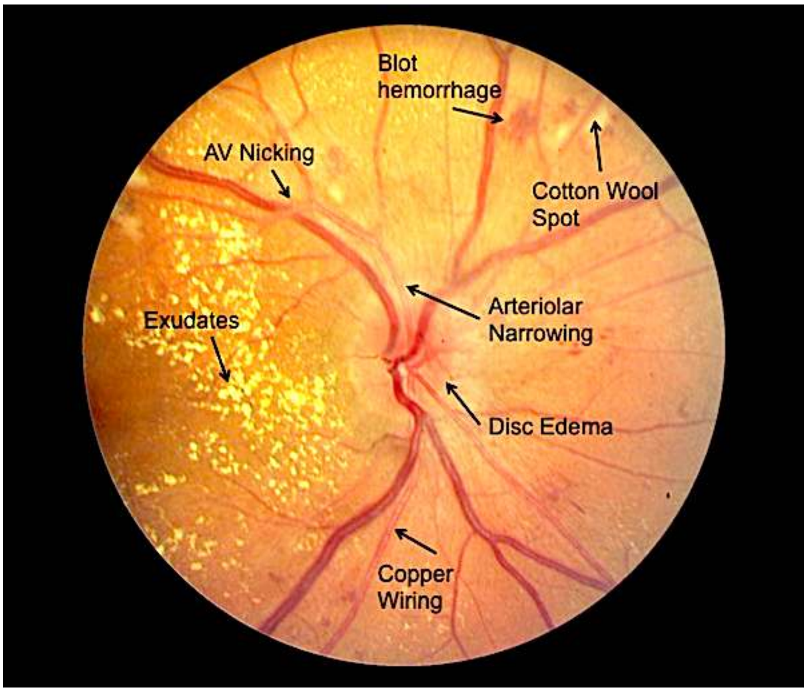
- Can present in numerous ways, such as with blurred or fluctuating vision, and difficulty recognizing colors.
- The USMLE vignette will usually say things like cotton wool spots (soft exudates), hard exudates, and retinal hemorrhages.
- Soft exudates are small, whitish lesions on the retina caused by microinfarctions and are axoplasmic material; hard exudates are lipoproteinaceous deposits due to leaky retinal vessels.
- Soft/hard exudates and retinal hemorrhages are seen classically in both diabetic and hypertensive retinopathies.



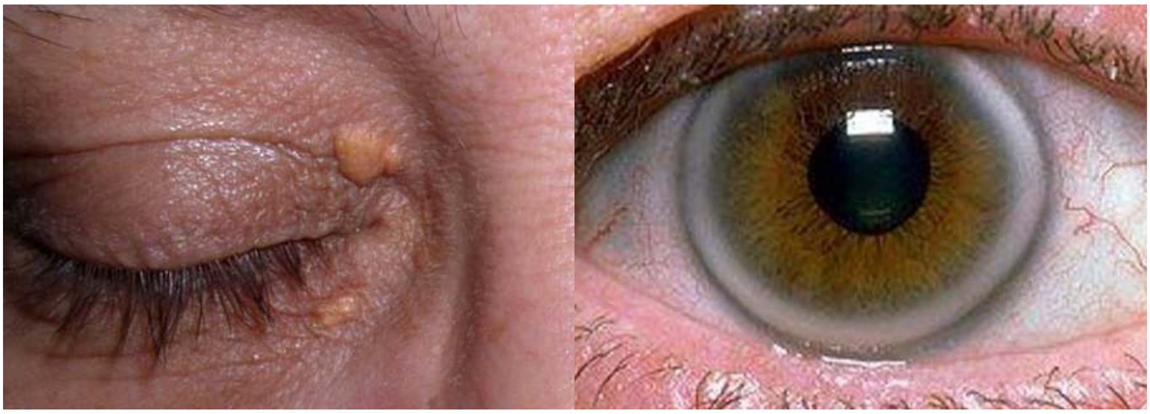


- Can be dry or wet (neovascular). Wet type can be treated with anti-VEGF agents.

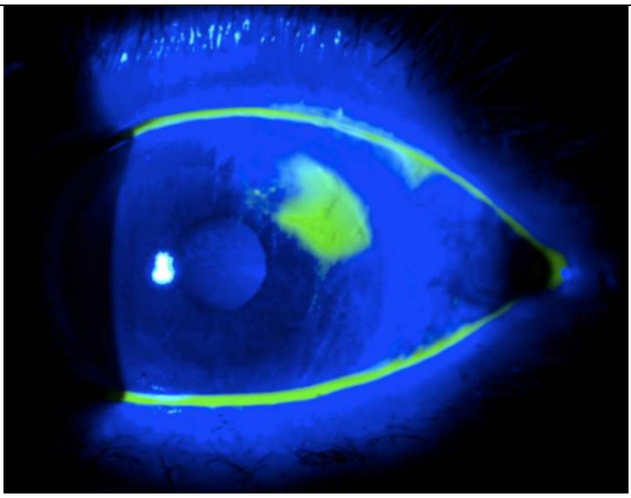
Hypertensive retinopathy
- Buzzy findings like AV nicking and arteriolar narrowing. - Q can mention "sausage-like" appearance of retinal arterioles.



Dyslipidemia
- Orbital xanthomata (cholesterol deposits; left image) and corneal arcus (lipid deposits; right image). - Both can occasionally be seen idiopathically in patients without lipid disorders (the latter in elderly).

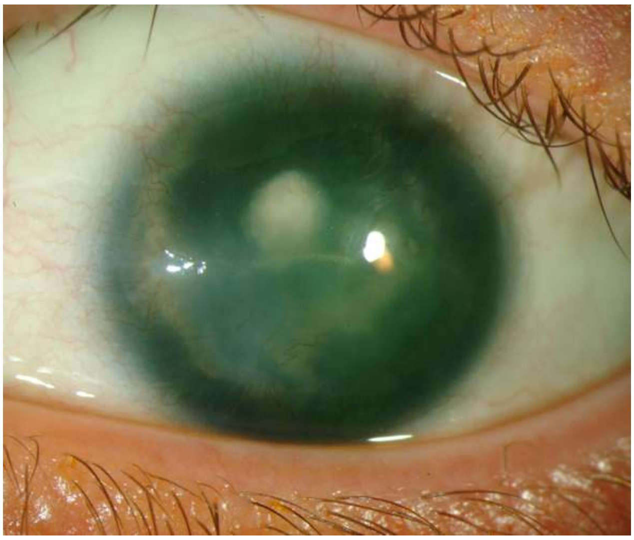
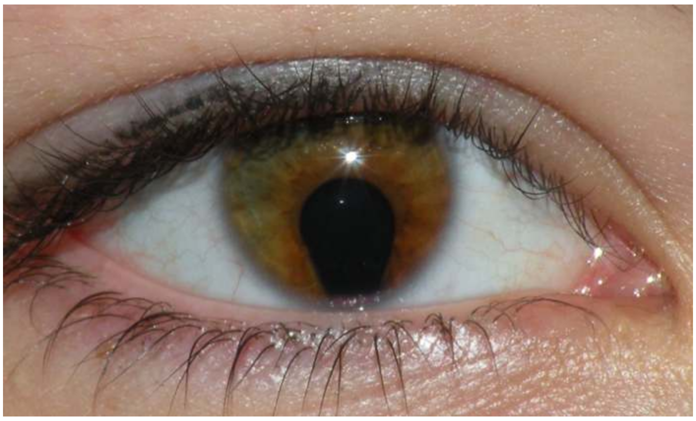


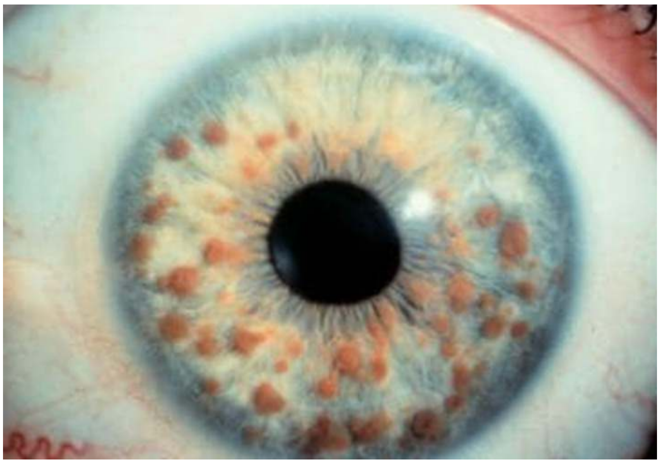
Corneal abrasion
- Scratchy or tearing sensation of the eye. Can be caused by contact lens-use or coming into contact with irritants such as sand (i.e., playing in sandbox with son/daughter). - USMLE will either ask for the next best step, where the answer is "instillation of fluorescein into the eye," or they will show you the following image (which is a fluorescein instillation) and then just ask for the diagnosis (i.e., corneal abrasion).



- Fluorescein is an orange dye that when dripped into the eye appears blue-green under violet light.
- Anything blue is normal. Anything green is abnormal.

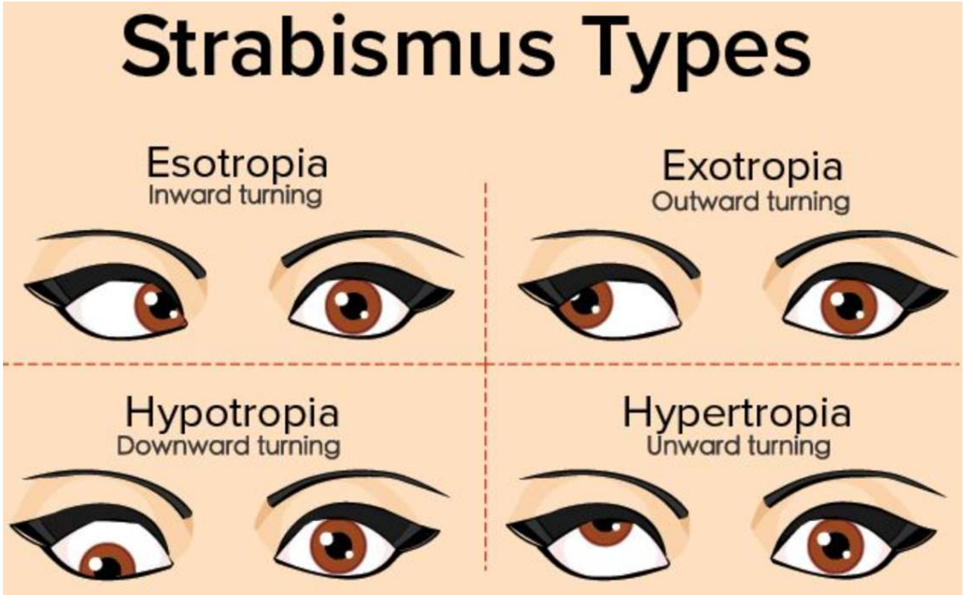
Subconjunctival hemorrhage
- Buzzy image / spot-diagnosis for USMLE. - Caused by bursting of superficial conjunctival vessels. Looks sinister but self-resolves uneventfully. - Can be caused by trauma or brief periods of increased pressure (i.e., severe vomiting or coughing). 

<table>
<tr><th colspan="2">Iris abnormalities</th></tr>
<tr><td>Aniridia</td><td>- Seen as part of WAGR complex with WT1 mutations – i.e., Wilms tumor, Aniridia, Genitourinary anomalies, Retardation.
</td></tr>
<tr><td>Coloboma</td><td>- Means hole in the eye. Seen in CHARGE syndrome, → Coloboma of the eye, Heart defects, Atresia of the choanae, Retardation of growth and development, Genitourinary anomalies, Ear defects.
</td></tr>
<tr><td>Albinism</td><td>- Can cause pale blue or pink irides (pleural for iris).
</td></tr>
</table>

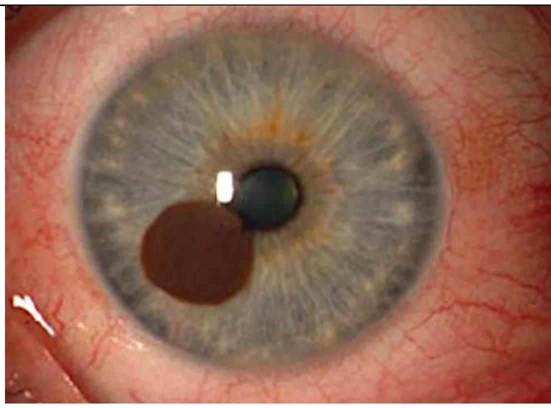
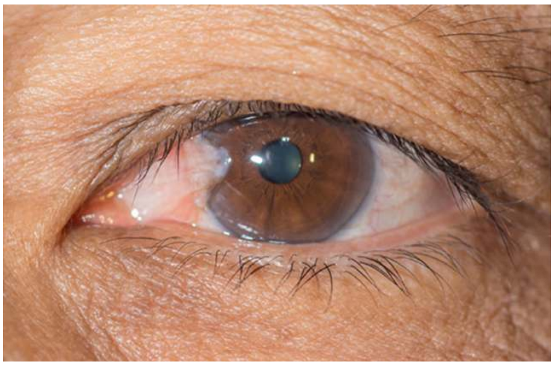
Lisch nodules	- Iris hamartomas seen in NF1.  - NF1 can present with axillary and groin freckling, café au lait spots (hyperpigmented macules), **optic glioma (tumor of CN II),** neurofibromas (nodules under the skin), pheochromocytoma, and weird brain cancers such as ependymoma and oligodendroglioma.
Brushfield spots	- Connective tissue deposits seen within the irides of some patients with Down syndrome. 

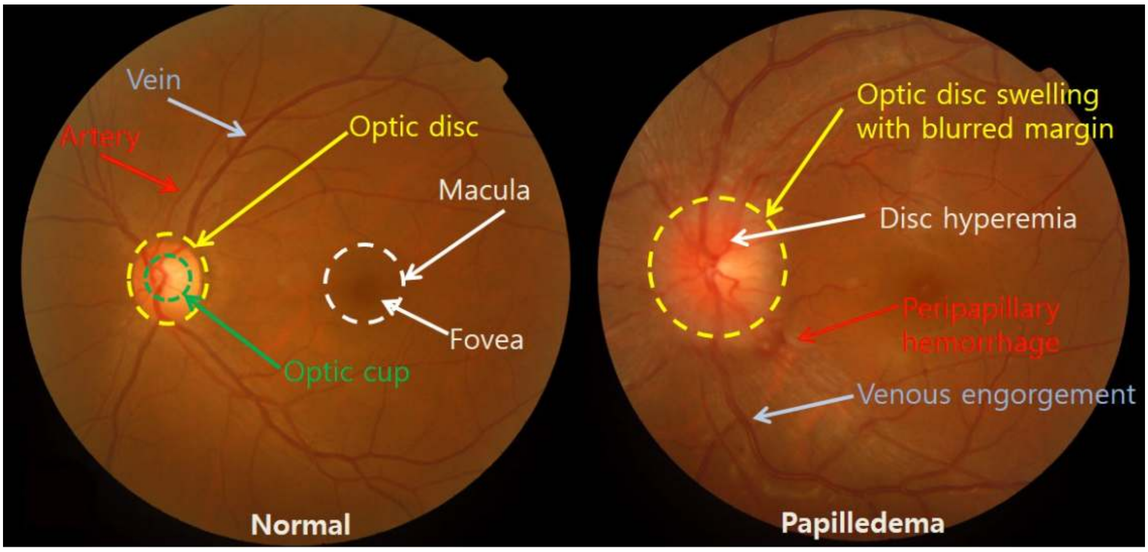
HY Peds Ophthal DDx for FM	
Orbital cellulitis	

	- Infection involving tissues posterior to orbital septum. - *Staph aureus* most common cause. - Rare sequela of adjacent infection, e.g., from the paranasal sinus. - Medical emergency + requires IV antibiotics. - Preseptal cellulitis is a less severe version of orbital cellulitis.
Hordeolum (stye)	- Painful bump on the eyelid. - *Staph aureus* infection of sweat gland or oil duct. - Treat with warm compresses. 
Chalazion	 - Painless bump on the eyelid. - Blocked oil duct; not an infection. - Ttreat with warm compresses.
Retinoblastoma	

	- The image shows leukocoria, which is an abnormal white appearance of the eye on light reflex (should normally appear red). - USMLE wants you to know that *RB* gene mutations not only cause congenital retinoblastoma, but also ↑ risk of osteosarcoma. - In other words, classic Q is they show you image above + ask what child is at risk for later → answer = osteosarcoma.
Strabismus	- Means "lazy eye," or misalignment of the eyes, due to weakened muscles in one eye.  - Can lead to **amblyopia,** which is reduced visual acuity in the weak eye due to the brain favoring the stronger eye during development. - Strabismus, however, is not reduced vision in the weak eye. It precedes amblyopia if not treated with eye exercises and/or penalization of (patching) the strong eye.

UV light-induced Ophthal pathologies	
Cataracts	- Progressive clouding or opacification of the lens, leading to blurred or hazy vision, particularly in bright light, with increased glare sensitivity.  - Can be caused by sunlight, but the most important cause on USMLE is diabetes mellitus, since high glucose entering the lens → converted to sorbitol via aldose reductase → sorbitol has high osmotic pull, causing osmotic damage to the lens and cataracts. - Recent changes in visual acuity can suggest new-onset or worsening diabetes. - Can be seen in Peds in congenital rubella, congenital syphilis, and NF2.
Melanoma	- Melanomas can grow from the conjunctiva and cornea. Just be aware they exist.

	
Pterygium	- Non-cancerous overgrowth of non-keratinized stratified squamous epithelium growing from the conjunctiva. - Crosses over the cornea (clear part of the eye covering the iris and pupil). - History of sun exposure is the major risk factor. 
Pinguecula	- Yellow-white benign lesion on the conjunctiva composed of protein and fat. - Does not cross over the cornea. - Sunlight is the major risk factor. 

<table>
<tr><th>Papilledema</th></tr>
<tr><td>- Means increased intracranial pressure on USMLE.
- Papilledema refers to blurring of the optic disc margins due inflammation of the optic nerve head in the setting of increased intracranial pressure.

</td></tr>
</table>

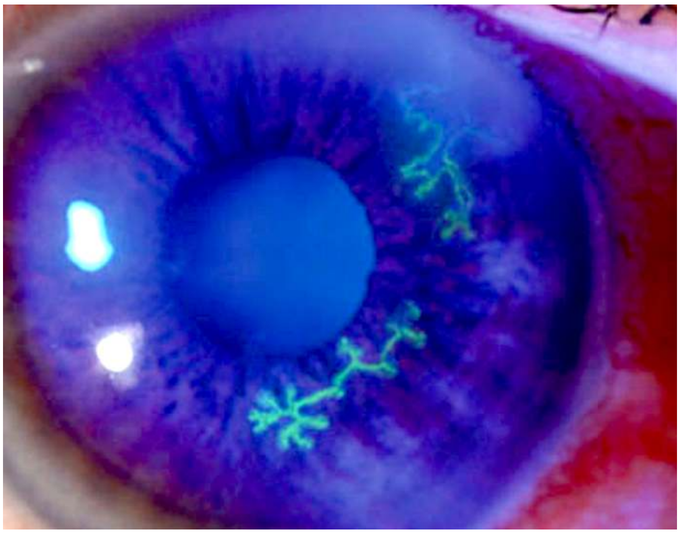
<table>
<tr><th colspan="2">HY Autoimmune eye pathologies</th></tr>
<tr><td>Anterior uveitis</td><td>- WBCs in the anterior chamber sometimes seen in flares of autoimmune diseases, especially sarcoidosis and ankylosing spondylitis.
- USMLE might sometimes just give you patient with sarcoidosis and then ask what needs to be done in terms of follow-up for the patient → answer = “slit-lamp examination” to check for anterior uveitis.</td></tr>
<tr><td>Keratoconjunctivitis sicca</td><td>- Dry eye (xerophthalmia) caused by decreased tear production.
- Classically autoimmune, either a stand-alone condition or as part of Sjögren syndrome.
- Latter presents additionally with dry mouth (xerostomia) and systemic findings such as arthritis.</td></tr>
<tr><td>Dermatomyositis</td><td>- Causes heliotrope rash (violaceous eyelids).
- Don’t confuse with the malar rash of lupus.

- Violaceous eyelids can also be seen in neuroblastoma in Peds, albeit completely unrelated. I just mention it cuz it shows up on an NBME form.</td></tr>
</table>

Miscellaneous hereditary conditions	
Leber hereditary optic neuropathy	- USMLE wants you to know that mitochondrial disorders, e.g., LHON, present as a triad of 1) eye/ear problems, 2) hypotonia, and 3) lactic acidosis (low bicarb; metabolic acidosis). - Other mitochondrial disorder names you could be aware of are MELAS (Mitochondrial Encephalopathy, Lactic Acidosis, and Stroke-like episodes) and MERRF (Myoclonic epilepsy with ragged red fibers). - Heteroplasmy is variability of phenotype severity across offspring (Step 1 term).
Osteogenesis imperfecta	- Collagen I mutation. - Can cause the very buzzy blue sclerae. 
Alport syndrome	- Mutation in type IV collagen; X-linked recessive. - The answer on USMLE for eye and/or ear problem + red urine.
Marfan syndrome	- Can cause lens dislocations.
Homocystinuria	- Can cause lens dislocations.

Ophthalmia neonatorum	
- Fancy way of saying neonatal conjunctivitis.	
Chlamydia	- Caused by *C. trachomatis* D-K (the STD). - Presents usually >1 week post-birth and can lead to **pneumonia** with a lymphocytic shift (since chlamydia is obligate intracellular, so requires lymphocytes for cell-mediated immunity to clear them, unlike extracellular bacteria which can be cleared humorally with neutrophils). - The chlamydia can drain through the nasolacrimal duct down to the lungs, causing the pneumonia. - Prophylaxis is treating the mom while pregnant if she has chlamydia; treatment in the neonate who already has the conjunctivitis is **oral** erythromycin. - NBME can give you vignette of wheezes, crackles, and fever in a 3-week-old who they say was treated for chlamydia conjunctivitis 2 weeks ago with topical erythromycin at birth. But this is not the correct treatment since topical will not cover any organism that has already entered the nasolacrimal duct.
Gonorrhea	- Gonococcal conjunctivitis will present in the first week of life with yellow discharge from the eye(s). - The USMLE vignette will be vague, where the student says, "But how are we supposed to know it's not chlamydia, just because it's first week of life, that's it?" And my response is: if they give you chlamydia ophthalmia neonatorum, they'll **always** tie it to subsequent pneumonia somehow, since that is such a HY point. - Prophylaxis for gonococcal is erythromycin ointment; treatment is IM 3rd-gen cephalosporin (usually cefotaxime in peds).
Chemical	- Chemical conjunctivitis is classically due to silver nitrate eyedrops that used to be given to neonates for gonococcal prophylaxis. However, they have been replaced by erythromycin ointment.

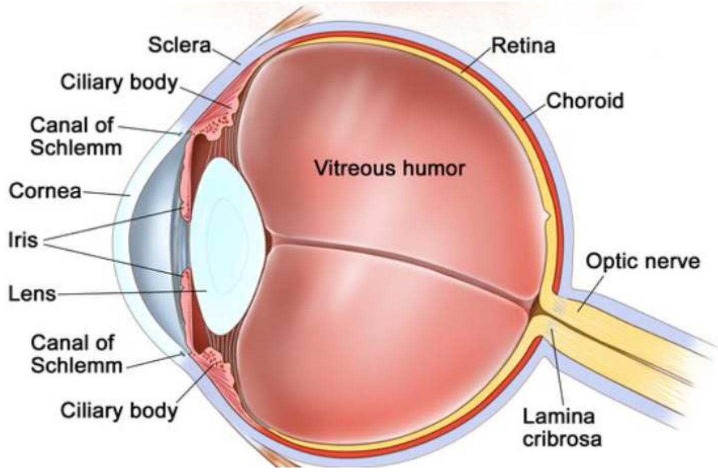
Viral	- Viral conjunctivitis will present as teary eye that is either unilateral or bilateral, and either itchy or non-itchy. I’ve seen variable presentation on USMLE. You just need to know this is the most likely diagnosis in both school-age kids and adults, and is caused by **adenovirus,** which is a DNA virus. Treatment is supportive with saline rinse.
Allergic	- USMLE wants you to know that “cobblestoning” of the tarsal conjunctiva is buzzy for allergic conjunctivitis. - Likewise, for allergic rhinitis, they can say cobblestoning of the nasal mucosa. - Some students have asked if allergic has to be bilateral, which it need not. And it doesn’t have to be itchy either.

Herpesviridae infections	
Herpes keratitis	- Keratitis is inflammation of the cornea. - HSV1/2 can cause dendritic (tree-like) pattern in the eye on fluorescein instillation.  - May or may not cause vesicles around the eye as well. - Tx with topical acyclovir/valacylcovir.
Herpes zoster ophthalmicus	- VZV (HHV-3) can cause shingles of the eye. - Causes keratitis with a similar dendritic pattern to HSV 1/2. - May or may not cause vesicles around the eye as well. - Tx with topical acyclovir/valacylcovir.
CMV retinitis	- The answer for blurry vision in severely immunocompromised patient (usually AIDS). - You don’t need to know the fundoscopy. - Treat with ganciclovir.

1) A 59-year-old man comes to the physician for a 6-month history of gradual peripheral visual loss in both eyes. He describes increasing “tunnel vision,” where his peripheral vision is significantly diminished and he can only see objects directly in front of him. He also experiences occasional eye pain, especially in low-light conditions. On examination, intraocular pressure (IOP) is 29 mg HH (NR 12-21). There is cupping of the optic discs. Which of the following is the most likely explanation for this patient’s findings?

A) Age-related macular degeneration
B) Cataracts
C) Diabetic retinopathy
D) Open-angle glaucoma
E) Retinal detachment

The answer is D (open-angle glaucoma).



Open-angle glaucoma is a chronic eye condition characterized by increased intraocular pressure due to impaired drainage of aqueous humor. Unlike acute angle-closure glaucoma, the drainage angle (canal of Schlemm) between the cornea and iris remains open. This condition progresses slowly and is usually painless. As it advances, it can cause peripheral vision loss (tunnel vision) and result in optic nerve damage characterized by "cupping of the optic disc" (i.e., an increased cup-to-disc ratio).

The USMLE wants **tonometry** as the first step in management (measures intraocular pressure).

Treatment for open-angle glaucoma for USMLE is not going to be a trivia game of "which drug first." They just want you to know that topical prostaglandins, such as **latanoprost,** can increase outflow of aqueous humor, thereby reducing IOP. Topical beta-blockers, such as **timolol,** decrease production of aqueous humor. **Carbonic anhydrase inhibitors,** such as dorzolamide, decrease production of aqueous humor. **Pilocarpine** is a muscarinic receptor agonist that constricts the pupil and increases outflow of aqueous humor.

Age-related macular degeneration presents with painless, gradual loss of central, not peripheral, vision. The USMLE might mention drusen, which is deposits of yellow lipoproteinaceous material on the retina.

Cataracts are a progressive clouding or opacification of the lens, leading to blurred or hazy vision, particularly in bright light, with increased glare sensitivity. The highest yield cause on USMLE is diabetes mellitus, since high glucose entering the lens → converted to sorbitol via aldose reductase → sorbitol has high osmotic pull, causing osmotic damage to the lens and cataracts. Recent changes in visual acuity can suggest new-onset or worsening diabetes.

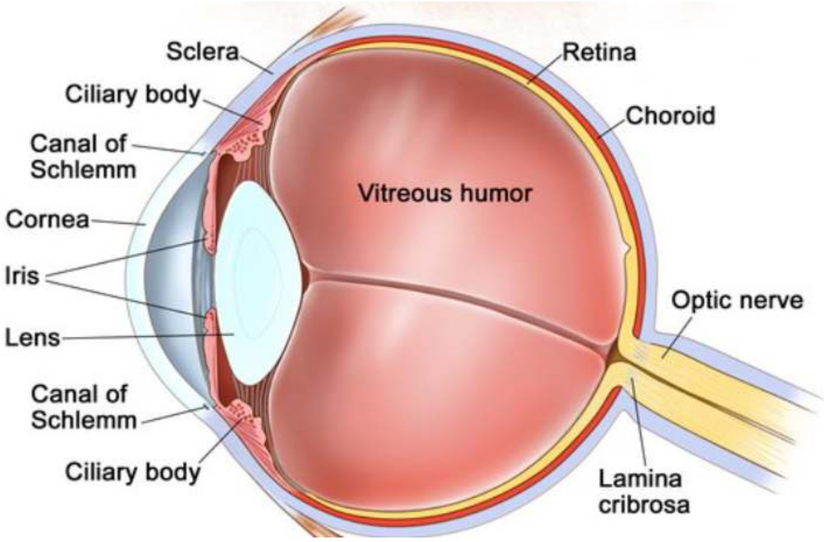
Diabetic retinopathy can present in numerous ways, such as with blurred or fluctuating vision, and difficulty recognizing colors. The USMLE vignette will usually say things like cotton wool spots (soft exudates), hard exudates, and retinal hemorrhages. Soft exudates are small, whitish lesions on the retina caused by microinfarctions and are axoplasmic material; hard exudates are lipoproteinaceous deposits due to leaky retinal vessels. Soft/hard exudates and retinal hemorrhages are seen classically in both diabetic and hypertensive retinopathies.

Retinal detachment will usually mention a boxer who got hit in the eye, or a patient who has Hx of severe myopia (a risk factor due to elongated shape of the eye). The fundoscopy is HY for USMLE and will show a well-demarcated lesion where the retina is protruding out into the vitreous.

2) A 61-year-old man presents to the ED with sudden-onset eye pain, headache, nausea, and blurred vision in his left eye. He describe seeing halos and colored rings of light. On examination, his left eye is red and teary, with a fixed mid-dilated pupil. Intraocular pressure is 30 mm Hg (NR 12-21). Which of the following is the most likely explanation for this patient's findings?

A) Cataracts
B) Closed-angle glaucoma
C) Conjunctivitis
D) Open-angle glaucoma
E) Retinal detachment

The answer is B (closed-angle glaucoma).



Closed-angle glaucoma on USMLE presents with a very buzzy red, teary, fixed mid-dilated pupil. It will be due to closure of the drainage angle (canal of Schlemm) between the cornea and the iris, leading to a rapid increase in IOP. This can lead to compression of the optic nerve and blood vessels, causing severe symptoms such as intense eye pain, headache, nausea, vomiting, blurred vision, and seeing colored rings or halos.

Similar to open-angle glaucoma, the USMLE wants **tonometry** as the first step to confirm increased IOP.

Treatment is IV mannitol (helps draw excess aqueous humor out of the eye), timolol, pilocarpine (muscarinic receptor agonist), or alpha-2 agonists such as brimonidine.

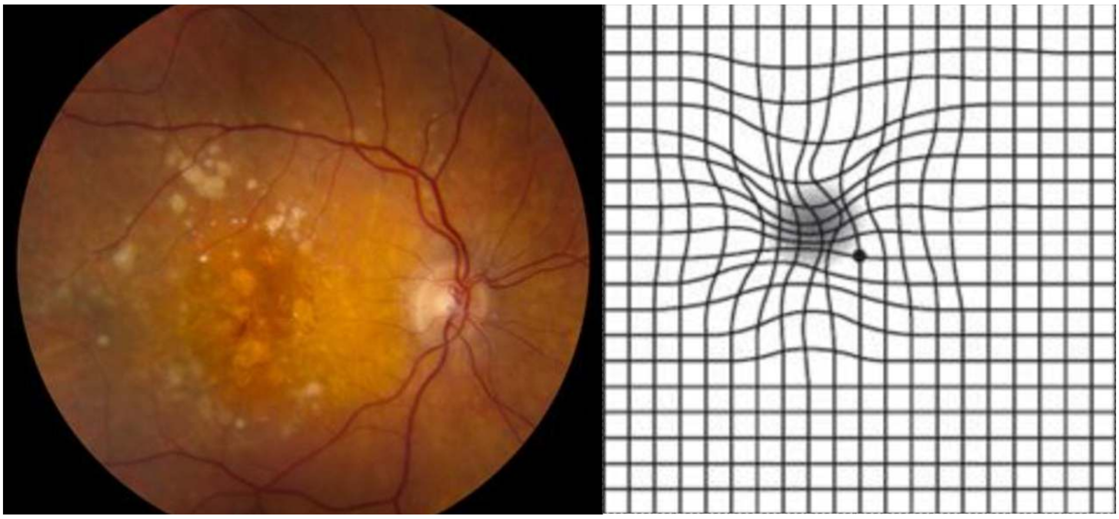
Cataracts are abnormal haziness of the lens. They will present as blurry vision in a patient who has diabetes or excessive exposure to sunlight (e.g., Hx of being a fisherman).

Conjunctivitis usually presents as redness of the eye that may or may not be itchy. There are many causes/types of conjunctivitis that I will talk about in this PDF.

Open-angle glaucoma presents as painless loss of peripheral vision over many months.

Retinal detachment presents as a “curtain coming down,” or acute loss of partial or total vision, usually in a patient who was hit in the eye or has Hx of severe myopia.

3) A 72-year-old man comes to the ophthalmologist for a follow-up appointment. Fundoscopic and Amsler grid results are shown below. Which of the following is the most likely explanation for these findings?

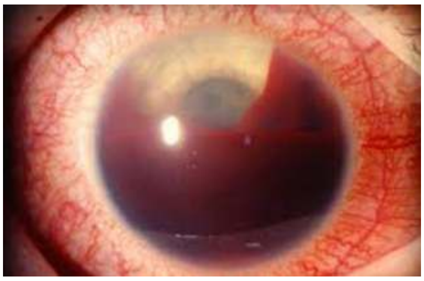


A) Hyphema
B) Macular degeneration
C) Open-angle glaucoma
D) Retinitis pigmentosa
E) Vitreous hemorrhage (convalescence stage)

The answer is B (macular degeneration).

USMLE wants you to know that "wavy lines" on a straight-line Amsler test = macular degeneration. It is characterized by loss of central vision. Fundoscopy shows drusen, which are yellow lipoproteinaceous deposits.

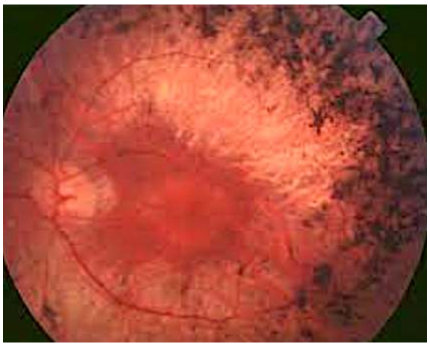
Hyphema is blood in the anterior chamber of the eye.



Hyphema

Open-angle glaucoma presents as painless loss of peripheral vision over many months.

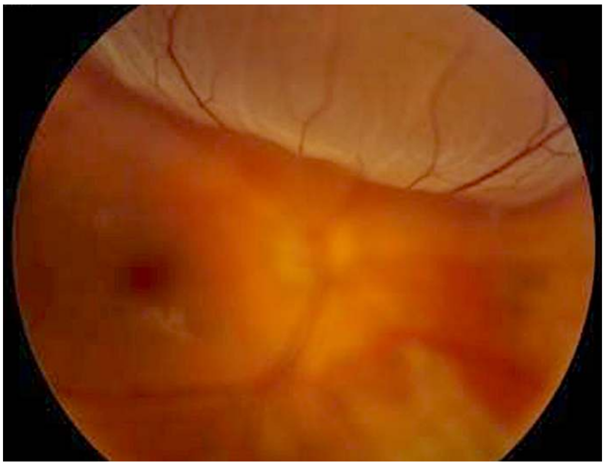
Retinitis pigmentosa presents as loss of peripheral vision and nyctalopia (loss of night vision). Fundoscopy will show pigment on the retina.



Retinitis pigmentosa

Vitreous hemorrhage (i.e., bleeding into the vitreous) will present as sudden-onset vision loss or floaters in someone with diabetes or hypertension.

4) A 54-year-old woman presents to emergency for a 2-hour history of painless loss of vision in her left eye following bumping into the corner of a cabinet. Examination shows visual acuity of 20/300 in the left eye and 20/30 in the right eye. She has history of hypertension and diabetes. Which of the following is the most likely diagnosis?



A) Central retinal artery occlusion
B) Central retinal vein occlusion
C) Hypertensive retinopathy
D) Retinal malposition
E) Vitreous detachment

The answer is D (retinal malposition).

The diagnosis is retinal detachment. It occurs classically in those with sudden trauma to the eye or severe myopia, as with this patient. The fundoscopy is HY and important for USMLE, especially 2CK.

Central retinal artery occlusion will present as sudden painless loss of vision in the eye in a patient who has Hx of hypertension, due to a carotid atheromatous plaque that's launched off to the eye. The retina can be described as pale with a cherry red spot; the latter is due to perfusion from the choroidal artery.

Central retinal vein occlusion will present with painless visual loss in a patient with diabetes or hypertension. Fundoscopy will show tortuous and dilated veins.

Hypertensive retinopathy will be described on USMLE as having arteriovenous nicking (AV nicking) and flame hemorrhages on fundoscopy. They can also mention soft exudates (cotton wool spots) and hard exudates. Once again, cotton wool spots are whitish, wispy axoplasmic material from microinfarcts. Hard exudates are lipoproteinaceous deposits from leaky retinal vessels.

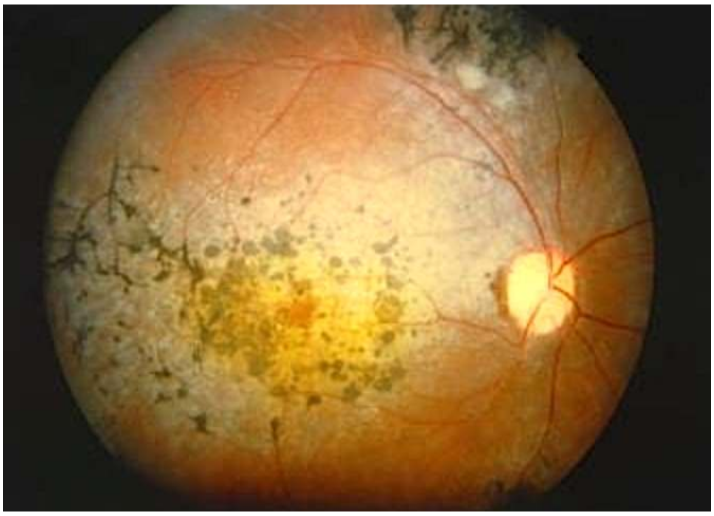
Vitreous detachment is a rare diagnosis that presents as sudden-onset floaters in one's field of vision. PVD occurs when the gel-like vitreous separates from the retina, usually due to age-related changes in vitreous consistency and structure. It is also a known rare sequela of LASIK eye surgery.

5) A 34-year-old woman presents to the physician for an 8-month history of decreased vision at night and worsening tunnel vision. Intraocular pressure is 18 mm Hg (NR 12-21). The patient's family history is notable for several relatives with similar visual symptoms. What is the most likely diagnosis?

A) Age-related macular degeneration
B) Diabetic retinopathy
C) Open-angle glaucoma
D) Presbyopia
E) Retinitis pigmentosa

The answer is E (retinitis pigmentosa).

Retinitis pigmentosa will present on USMLE as a combo of nyctalopia (loss of night vision) and loss of peripheral vision. There will often be a family history. Fundoscopy shows characteristic pigmentation.



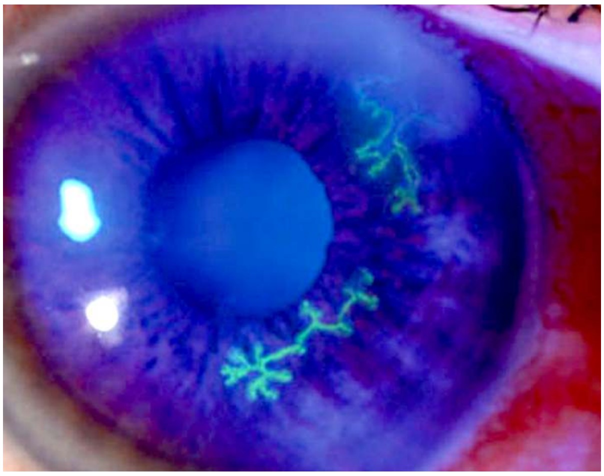
Age-related macular degeneration will present as loss of central vision. The USMLE can mention wavy lines on an Amsler grid test. Drusen will be seen on fundoscopy, which is shown as yellow lipoproteinaceous deposits. Even though hard exudates are also lipoproteinaceous deposits, the latter are due to leaky retinal vessels in diabetes and hypertension.

Diabetic retinopathy will present with cotton wool spots, hard exudates, and hemorrhages on the fundoscopy in a patient with advanced diabetes. Changes in vision can be multifarious.

Open-angle glaucoma presents with painless loss of peripheral vision over many months to years. But IOP is elevated, not normal, in glaucoma. Glaucoma, by definition, is increased IOP.

Presbyopia is age-related loss of the eye's ability to focus on near objects due to the lens losing flexibility with time.

6) A 39-year-old woman comes to the physician for a 4-day history of a painful, tearing right eye. Vitals are within normal limits. Past medical history is unremarkable. A fluorescein instillation of the eye is shown. The most likely causal organism is most taxonomically similar to which of the following?



A) Echovirus
B) Hepatitis A
C) HIV
D) Measles virus
E) Parvovirus B19
F) West Nile virus

The answer is E (Parvovirus B19).

The diagnosis is likely HSV keratitis, which means inflammation of the cornea. It presents as a dendritic (tree-like) pattern in the eye on fluorescein instillation. VZV (shingles) can also cause keratitis with a similar pattern. Both HSV1/2 and VZV (HHV-3) are Herpesviridae, which are DNA viruses. Parvo is the only DNA virus listed in the answer choices.

Echovirus is RNA and is the most common cause of viral (aseptic) meningitis. CSF will show high lymphocytes, normal glucose, and normal or slightly-elevated protein.

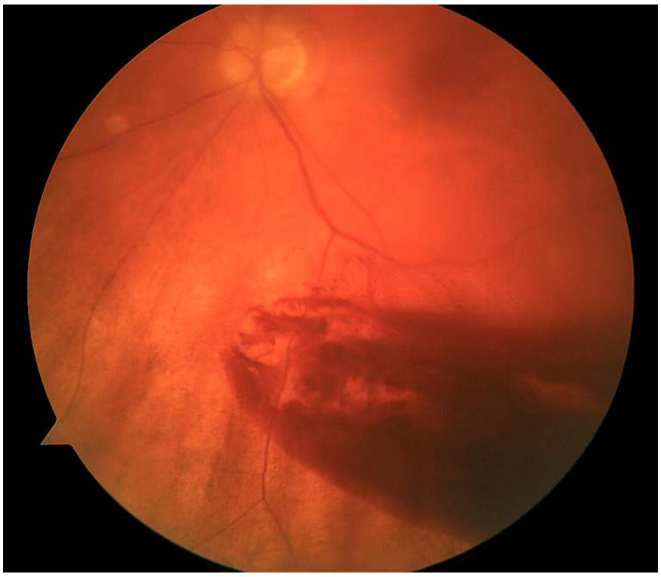
Hepatitis A and E are both RNA viruses that cause acute hepatitis. HepA is the most common acute hepatitis in the USA and Mexico. Hepatitis E is more Asia / Tibet and causes fulminant hepatitis in pregnant women with high chance of death.

HIV is an RNA virus. USMEL wants you to know that highly-active anti-retroviral therapy (HAART) is always 3 drugs, and you do not commence TMP/SMX prophylaxis for PJP until the CD4 count drops below 200.

Measles is an RNA virus. It causes a head-to-toe maculopapular rash, often with Koplik spots on the buccal mucosa.

An offline NBME form for Step 1 assesses West Nile virus as that which is spread by *Culex* mosquito and has a bird reservoir. It is an RNA virus.

7) A 71-year-old woman comes to the physician for a 12-hour history of pain and blurry vision in her left eye accompanied by floaters and flashing lights. This began when she was walking in the wind from her car to the front door of her house. She has type II diabetes managed with insulin and hypertension managed with lisinopril. Which of the following is the most likely diagnosis?



A) Central retinal artery occlusion
B) Central retinal vein occlusion
C) Corneal abrasion
D) Hordeolum
E) Vitreous hemorrhage

The answer is E (vitreous hemorrhage).

Vitreous hemorrhage is diagnosis of exclusion, meaning we eliminate to get there. The USMLE does not expect you to know this fundoscopy. The point of this question is for you to merely know that the biggest risk factors of bleeding into the vitreous are diabetes and hypertension.

Central retinal artery occlusion will present as sudden, painless vision loss in a patient where a “curtain comes down.” This will be due to a carotid atheroma that’s launched off to the retinal vessels. A pale retina will be seen on fundoscopy with a cherry red spot due to maintenance of perfusion from the choroidal artery.

Central retinal vein occlusion will be described as tortuous veins on fundoscopy in an older patient with visual disturbance with or without pain. Diabetes and hypertension are risk factors.

Corneal abrasion will be present as a heavily teary eye with an intensely scratchy sensation. USMLE will give you a guy who works in a foundry grinding metal or who was playing in the sandbox with his kid. Fluorescein instillation of the eye will show a blue eye with an irregular green area. A 2CK NBME Q might give you a simple vignette of corneal abrasion and then answer is just “instillation of fluorescein” as next best step.

Hordeolum (aka sty) is a painful bump on the eyelid that is typically due to *S. aureus* infection. Warm compresses are next best step.

8) A 1-year-old boy is brought to the physician by his parents after they noticed abnormal coloration between his eyes in some light. This patient is at greatest risk for development of which of the following?

A) Clusters of mucin-producing glandular neoplastic cells
B) Lacunae filled with neoplastic chondrocytes
C) Neoplastic sheets of plasma cells
D) Pleomorphic neoplastic cells with production of new woven bone
E) Tightly packed uniform, small, round blue cells

The answer is D (pleomorphic neoplastic cells with production of new woven bone).

Choice D refers to osteosarcoma, which patients with hereditary retinoblastoma have increased risk for developing later in life. The child has leukocoria, which is an abnormal white appearance of the eye on the light reflex (should normally appear red). *RB* is a tumor suppressor gene (i.e., two hits are required for cancer to develop) but is considered autosomal dominant because almost all patients with one mutation go on to get the disease (since a second mutation will occur within first year of life).

The three causes of osteosarcoma are: 1) *Rb* mutations; 2) idiopathic in males 15-35; and 3) in elderly patients who have Paget disease of bone.

"Clusters of mucin-producing glandular neoplastic cells" refers to signet-ring cells, which are gastric cancer. They can also be Krukenberg tumors (gastric cancer that has metastasized bilaterally to the ovaries).

"Lacunae filled with neoplastic chondrocytes" is the histo description for chondrosarcoma, which will be described as a "glistening" tumor grossly. I discuss bone tumors in more detail in my Anatomy/MSK PDF.

"Neoplastic sheets of plasma cells" refers to multiple myeloma. MM is a cancer of plasma cells, where the bone marrow will have >10% plasma cells, which can be described as having "clock face chromatin" on a blood smear. The first step in diagnosis for multiple myeloma is serum protein electrophoresis (HY for 2CK) showing a kappa or lambda IgG light-chain spike (aka M-protein spike).

"Tightly packed uniform, small, round blue cells" refers to Ewing sarcoma, which is the answer for bone tumor in pediatrics almost always. It can present with fever and pain, making it appear similar to osteomyelitis. The difference, however, is the Ewing has increased uptake in the diaphysis on bone scan, whereas osteomyelitis has increased uptake in the metaphysis. Ewings can be caused by an t(11;22) translocation and have "onion skinning" on histo.

9) During the past month, a 65-year-old man has had 6 episodes of blindness in his right eye lasting 10-20 minutes. Past medical history is noncontributory. He takes no medications. Which of the following is the most appropriate next step in diagnosis?

   A) Carotid duplex scan
   B) Carotid angiography
   C) 24-hour ECG monitor
   D) CT scan of the head
   E) Tonometry

The answer is A (carotid duplex scan).

The diagnosis is amaurosis fugax (i.e., painless, temporary loss of vision) due to atheromatous emboli from the common carotid arteries launching off to the retinal arteries.

If the NBME gives you amaurosis fugax in a patient with no past medical history or with hypertension, you must first assume carotid stenosis due to atherosclerosis as the cause. Hypertension is the most common cause of atherosclerosis of the carotids due to endothelial damage from the systolic impulse.

Carotid angiography I've never seen as a correct answer on NBME exams. It could theoretically be done after carotid duplex ultrasound to more definitively assess degree of carotid stenosis.

24-hour ECG monitor is the answer for patients with unexplained syncope (i.e., fainting) who have no past medical history. This is in order to look for arrhythmia (i.e., usually atrial fibrillation) as the cause of the syncope. It can also be done in patients who have stroke, TIA, or retinal artery occlusion if the vignette gives you a normotensive patient who's over the age of 75 (i.e., much more likely to be atrial fibrillation causing a left-atrial mural thrombus to launch off).

Non-contrast CT of the head could be done in the setting of TIA or stroke looking for bleeds.

Tonometry is a pressure study of the eye. It is first step on NBME for suspected glaucoma.

10) A 39-year-old woman comes to the physician for a 4-day history of blurry vision and eye pain on the left. Vitals are within normal limits. She is enrolled in a methadone program. Which of the following is the most likely explanation for this patient's presentation?

A) DNA virus; enveloped; circular
B) DNA virus; enveloped; linear
C) DNA virus; non-enveloped; circular
D) RNA virus; double-stranded; segmented
E) RNA virus; helical; segmented

The answer is B (DNA virus; enveloped; linear).

The diagnosis is CMV retinitis. CMV is in the Herpesviridae family, which are all DNA, enveloped, and linear. Even though the Step 1 is pass/fail now, this specific structural point about the Herpesviridae is HY.

The methadone program implies the patient might be HIV positive, since methadone is an opioid-receptor agonist used for heroin and opioid withdrawal.

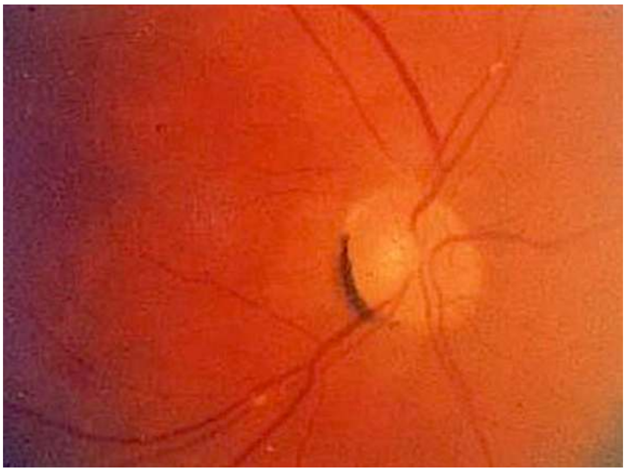
"DNA; enveloped; circular" is hepatitis B.

"DNA virus; non-enveloped; circular" is HPV and JC/BK polyoma viruses.

"RNA virus; double-stranded; segmented" is rotavirus.

"RNA virus; helical; segmented" is most notably influenza virus.

11) A 68-year-old woman comes to the physician 12 hours after an episode of blindness in her left eye that lasted 15 minutes. She takes no medications. Blood pressure is 138/80. Motor and sensory examination are normal. Pupils are equal, round, and reactive to light. Visual fields are full. Ocular movements are intact. Funduscopic examination shows a refractile body. Which of the following is the most appropriate next best step in diagnosis?



A) CT angiography
B) Slit-lamp examination
C) Tonometry
D) Transesophageal echocardiography
E) Ultrasound

The answer is E (ultrasound).

Same as with a couple questions earlier, carotid duplex ultrasound needs to be done in patients who have amaurosis fugax in order to look for carotid atherosclerosis.

“Refractile body” is a buzzy term the NBME can use to describe an embolus visualized in the eye.

CT angiography can be done to look for definitively for degree of occlusion of the carotids. I’ve never seen this as an answer on NBME for this purpose.

Slit-lamp examination is the answer when looking for anterior uveitis in autoimmune disease, such as ankylosing spondylitis. Anterior uveitis means inflammation of the iris and ciliary body. The ciliary body is smooth muscle that controls the shape of the lens. Slit-lamp exam is also the answer when looking for Kayser-Fleischer rings in Wilson disease.

Tonometry is a pressure study of the eye that is the first answer when diagnosing glaucoma.

Transesophageal echocardiography is done to confirm the presence of a left-atrial mural thrombus in a patient with known atrial fibrillation.

12) A previously healthy 5-year-old girl is brought to the physician because of itching of her left eye for the past 2 weeks. Her vitals are within normal limits. Examination shows conjunctival erythema and cobblestoning. The nasal mucosa is boggy and pale. She has been engaged in arts and crafts activities at school the past month. Which of the following is the most likely explanation for this patient's presentation?

A) Allergic conjunctivitis
B) Chemical conjunctivitis
C) Chlamydial conjunctivitis
D) Gonococcal conjunctivitis
E) Kawasaki disease
F) Viral conjunctivitis

The answer is A (allergic conjunctivitis).

USMLE wants you to know that “cobblestoning” is buzzy for allergic conjunctivitis and rhinitis on USMLE (the latter, i.e., if they say cobblestoning of the nasal mucosa). Some students have asked if allergic has to be bilateral, which it need not. And it doesn’t have to be itchy either.

Chemical conjunctivitis is classically due to silver nitrate eyedrops which used to be given to neonates for gonococcal prophylaxis, however they have been replaced by erythromycin ointment.

Chlamydial conjunctivitis is caused by *C. trachomatis* D-K (the STD). It presents usually >1 week post-birth and can lead to **pneumonia** with a lymphocytic shift (since chlamydia is obligate intracellular, so requires lymphocytes for cell-mediated immunity to clear them, unlike extracellular bacteria which can be cleared humorally with neutrophils). The chlamydia can drain through the nasolacrimal duct down to the lungs, causing the pneumonia.

Prophylaxis is treating the mom while pregnant if she has chlamydia; treatment in the neonate who already has the conjunctivitis is **oral** erythromycin.

NBME can give you vignette of wheezes, crackles, and fever in a 3-week-old who they say was treated for chlamydia conjunctivitis 2 weeks ago with topical erythromycin at birth. But this is not the correct treatment since topical will not cover any organism that has already entered the nasolacrimal duct.

Gonococcal conjunctivitis will present in the first week of life with yellow discharge from the eye(s). The USMLE vignette will be vague, where the student says, “But how are we supposed to know it’s not chlamydia, just because it’s first week of life, that’s it?” And my response is: if they give you chlamydia ophthalmia neonatorum (neonatal conjunctivitis), they’ll **always** tie it to subsequent pneumonia somehow, since that is such a HY point.

Prophylaxis for gonococcal is erythromycin ointment; treatment is IM 3rd-gen cephalosporin (usually cefotaxime in peds).

Kawasaki disease is a medium-large-vessel vasculitis in peds that presents with **5+ days of fever**, desquamation of the palms and soles, edema of the dorsa of the hands, cervical lymphadenopathy, and injection (redness) of the lips, tongue, and/or eyes. Students obsess over coronary aneurysms, but they’re actually LY. I’d say the highest yield points are the **5+ days** of fever and palms/soles desquamation. Treatment is with aspirin and IVIG. Kawasaki is the only thing you can give aspirin for in peds. Never give aspirin otherwise (can cause Reye syndrome).

Viral conjunctivitis will present as teary eye that is either unilateral or bilateral, and either itchy or non-itchy. I’ve seen variable presentation on USMLE. You just need to know this is the most likely diagnosis in both school-age kids and adults, and is caused by **adenovirus,** which is a DNA virus. Treatment is supportive with saline rinse.

13) A 64-year-old man with type II diabetes mellitus and hypertension comes to the physician for a routine ophthalmologic examination. He reports no visual disturbances. Fundoscopic examination shows a refractile body in the right eye. He works in metallurgy. Which of the following is the most likely cause of these findings?

A) AV nicking
B) Cholesterol
C) Cotton wool spot
D) Drusen
E) Foreign body
F) Hard exudate

The answer is B (cholesterol).

The patient likely has carotid stenosis due to hypertension that has led to an atheromatous plaque launching off to the eye. I've seen the NBME use the phrase "refractile body" to refer to an embolus observed fundoscopically.

AV (arteriovenous) nicking is a buzzy term that is a fundoscopic finding in hypertensive retinopathy. USMLE doesn't expect you to be able to identify this visually, don't worry. They'll say it sometimes in a stem.

Cotton wool spots (soft exudates) are white, wispy, axoplasmic material due to microinfarcts within retinal vessels. They are seen in hypertension and diabetes.

Drusen is a yellow, lipoproteinaceous material seen as deposits on the retina in age-related macular degeneration.

Foreign body penetration into the eye will cause a stinging and teary eye, e.g., in a foundry worker who grinds metal. USMLE wants "instillation of fluorescein" into the eye as the next best step.

Hard exudates are lipoproteinaceous deposits seen on the retina in hypertension and diabetes.

14) A 37-year-old woman comes to the physician for a 1-week history of blurry vision in the left eye and a repeated need to run to the bathroom approximately 12 times per day. Neurologic examination shows clonus of the right hand. Which of the following is most likely to be seen in this patient?

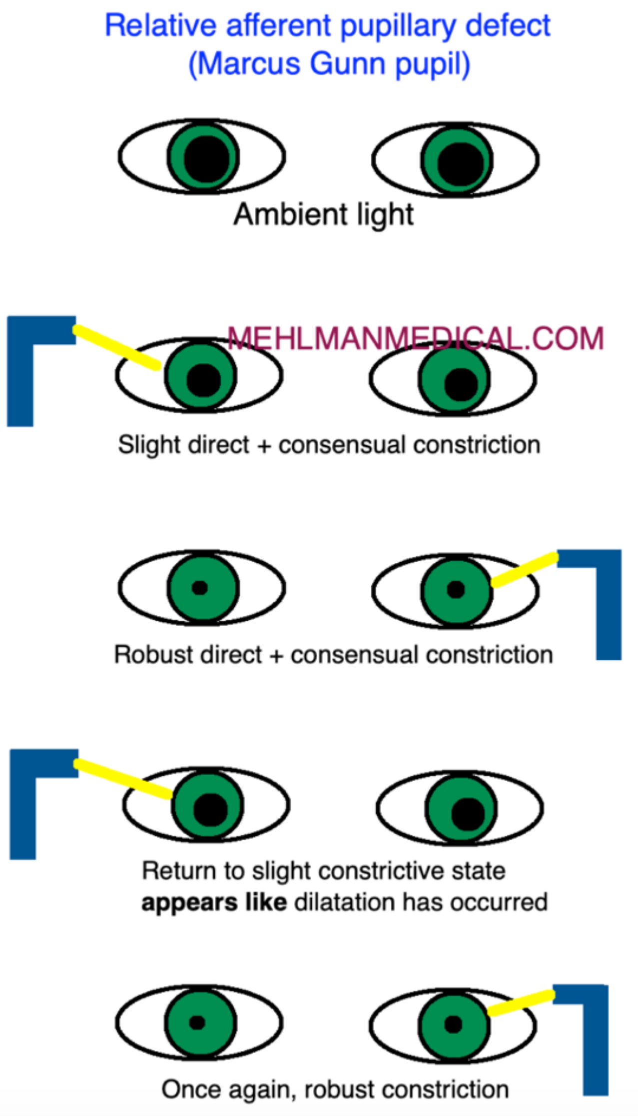
   A) Carotid atherosclerosis
   B) Endophthalmitis
   C) Hypertensive retinal artery changes
   D) Left atrial mural thrombus
   E) Optic neuritis

The answer is E (optic neuritis).

Diagnosis is multiple sclerosis, which is classically seen in women 20s-30s. The patient has urge incontinence and an upper motor neuron finding (clonus). Urge incontinence is exceedingly HY for MS, as are miscellaneous UMN findings such as Babinski sign. Diagnosis is with MRI showing white matter plaques in the CNS.

There are two main eye findings in MS: optic neuritis and internuclear ophthalmoplegia (discussed later).

Optic neuritis is inflammation of CN II. This can cause a Marcus Gunn pupil (aka relative afferent pupillary defect). In this case, the intensity of the light is under-sensed by the affected eye (afferent signal), so the resultant efferent CN III (oculomotor) parasympathetic output (back to the eyes for pupillary constriction) is attenuated; this means when you shine a light in the affected eye, both pupils constrict less in comparison to when the light is shone in the unaffected eye; this **gives the mere impression** that the pupils dilate when light is shone in the affected eye. Still confusing? Take a look at the following illustration (assuming a lesion of the patient's right eye [left side of illustration]):



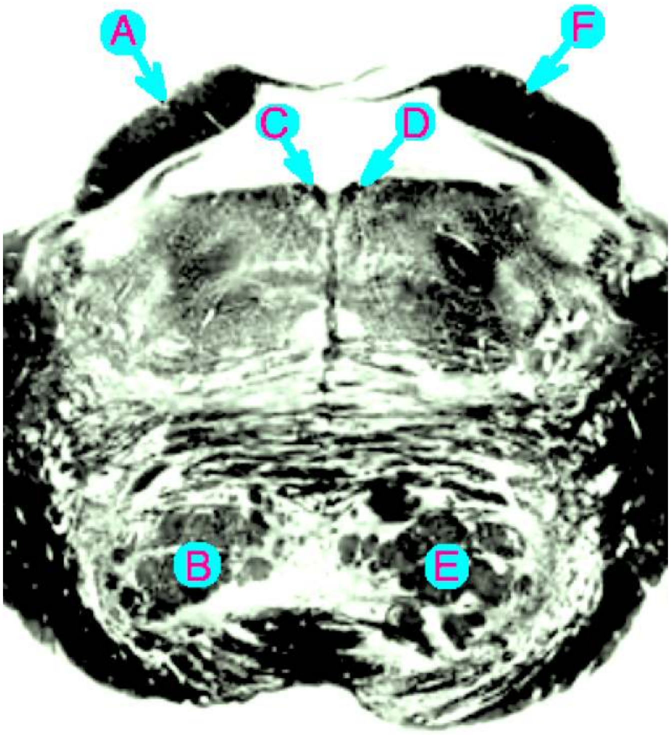
Carotid atherosclerosis would be seen generally in a patient over 50 with hypertension, diabetes, and/or smoking history, and would lead to retinal artery occlusion(s), where a curtain comes down over the visual field. A "refractile body" might be seen on fundoscopy.

Endophthalmitis is a deep infection of the eye that can occur with penetrating trauma or as a post-operative complication.

Hypertensive retinal artery changes would be described with buzzy findings such as: AV nicking; arteriolar narrowing; "sausage-like" appearance of retinal arterioles with narrowing.

Left-atrial mural thrombus would occur in the setting of AF, where it can launch off to the eye causing retinal artery occlusion(s). The vignette will usually be a normotensive patient over 70-75 (hypertensive / younger usually implies carotid stenosis). AF can also be seen in conditions such as hyperthyroidism.

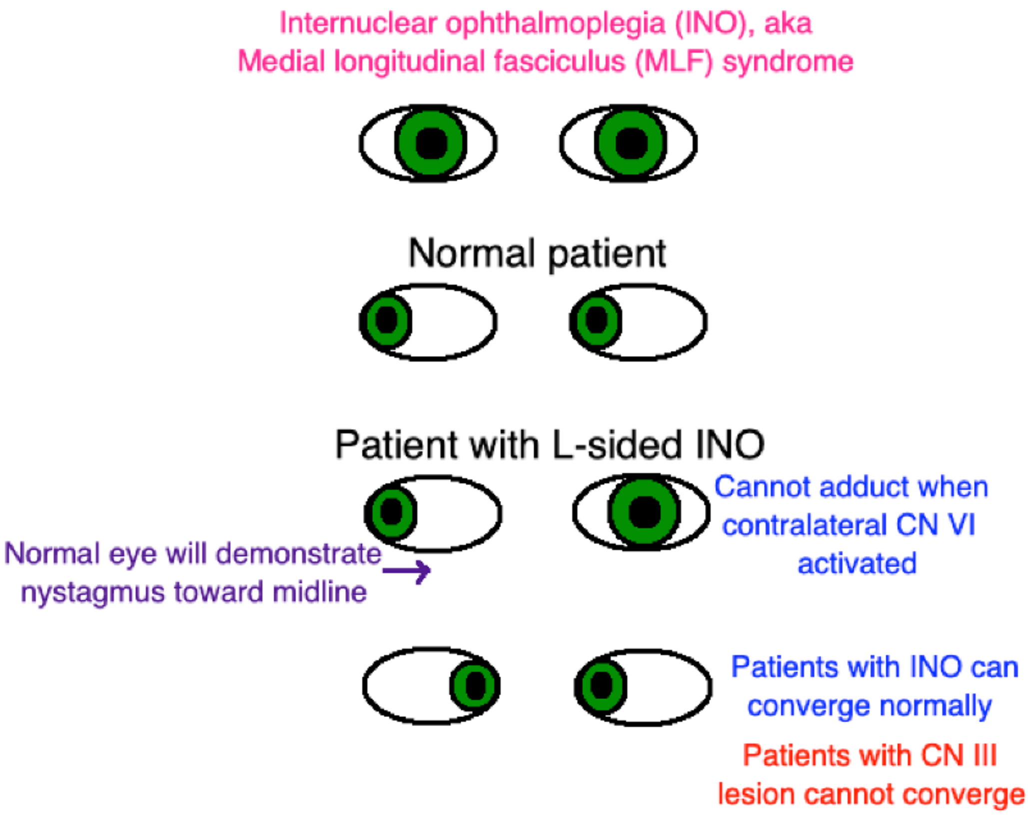
15) A 32-year-old woman comes to the physician for a 5-day history of visual disturbance. Ophthalmologic examination shows that when she looks to the left, the left eye demonstrates nystagmus and the right eye looks straight ahead. When she looks to the right, the left eye adducts and the right eye abducts. Convergence is normal. Which of the following areas is the most likely source of her pathology?



The answer is C (right-sided MLF).

Diagnosis is medial longitudinal fasciculus syndrome (MLF syndrome; aka internuclear ophthalmoplegia; INO) in a patient with multiple sclerosis (vignette also describes urge incontinence); normal convergence tells us that CN III is intact bilaterally.

What you need to know is this: in INO, the side that cannot adduct is the side that's fucked up, so we know all letters on the patient's left (D-F) are wrong; so looking at A-C, even if you have no idea, if you recognize the presentation as MLF syndrome, you can say, "Well, the name of the condition is **medial** longitudinal fasciculus syndrome, so I'll just choose the letter that's **medial.**"



Mechanism for pathology: when you look to one side (let's say the right), CN VI on the right and CN III on the left are activated. This unison of activation is accomplished when the left MLF is activated. In the case of MLF syndrome, the right eye can abduct without a problem, but the left eye will fail to adduct and the right eye will exhibit nystagmus back toward the midline. **MLF syndrome is pathognomonic for multiple sclerosis.** MS patients can also get optic neuritis (i.e., change in visual acuity, color vision, central scotoma, etc.), but this is not pathognomonic for MS (i.e., it can be seen in other circumstances, such as with ethambutol or sildenafil use).

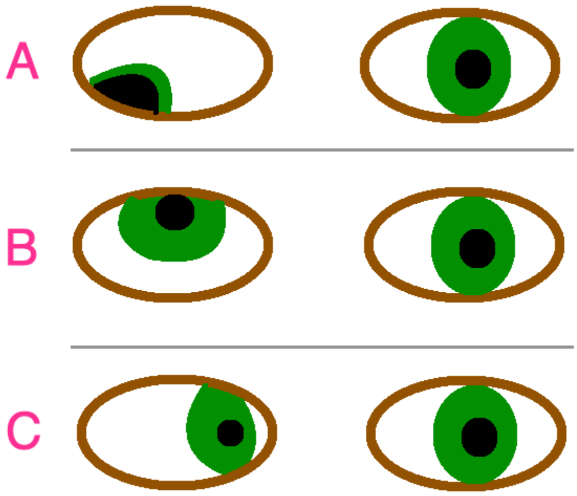
Exceedingly rare but MLF syndrome may occur in stroke due to HTN (still considered pathognomonic for MS).

Don't worry about the other lettered answer choices. A/F are inferior colliculi. B/E are pyramidal tracts.

16) A 42-year-old man is admitted to hospital following a car accident in which he sustained head trauma. On physical examination, his left eye is in a down and out position. Which of the following is the most likely location of his injury?

A) Abducens nerve
B) Oculomotor nerve
C) Optic nerve
D) Medial longitudinal fasciculus
E) Trochlear nerve

The answer is B (oculomotor nerve).



A = lesion of CN III (oculomotor) → presents as down and out; B = CN IV lesion (trochlear) → presents as upward gaze + patient will tilt head toward unaffected side; C = CN VI (abducens) → medially directed eye.

PCoM aneurysmal compression of CN III affects superficial parasympathetic efferents leading to ipsilateral mydriasis and loss of accommodation; diabetes may affect deep extraocular muscle efferents leading to partial ptosis and down-and-out pupil.

In other words, you need to know for USMLE that a PCOM aneurysm can cause an ipsilateral blown pupil.

MLF syndrome was discussed in the prior Q. It would lead to failure to adduct one of the eyes when looking to one side.

An optic nerve lesion would cause optic neuritis (blurry vision, change in color vision, central scotoma, etc.), or just outright blindness.

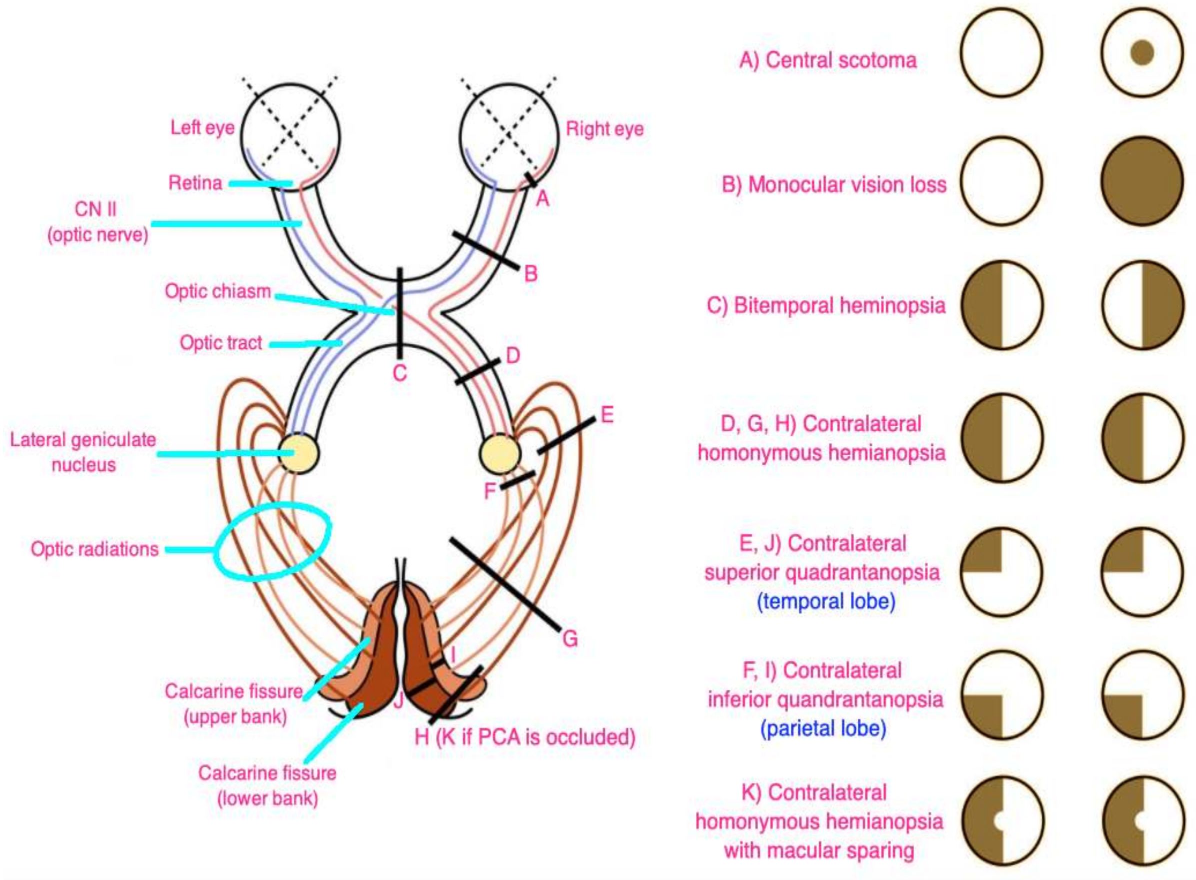
17) A 29-year-old man comes to the physician for a 2-month history of bumping into walls while walking around corners. He has a 6-month history of occasional serous discharge from both nipples. Which of the following structures is most likely experiencing impingement in this patient?

A) Occipital lobe
B) Optic chiasm
C) Optic radiation
D) Optic tract
E) Retina

The answer is B (optic chiasm).

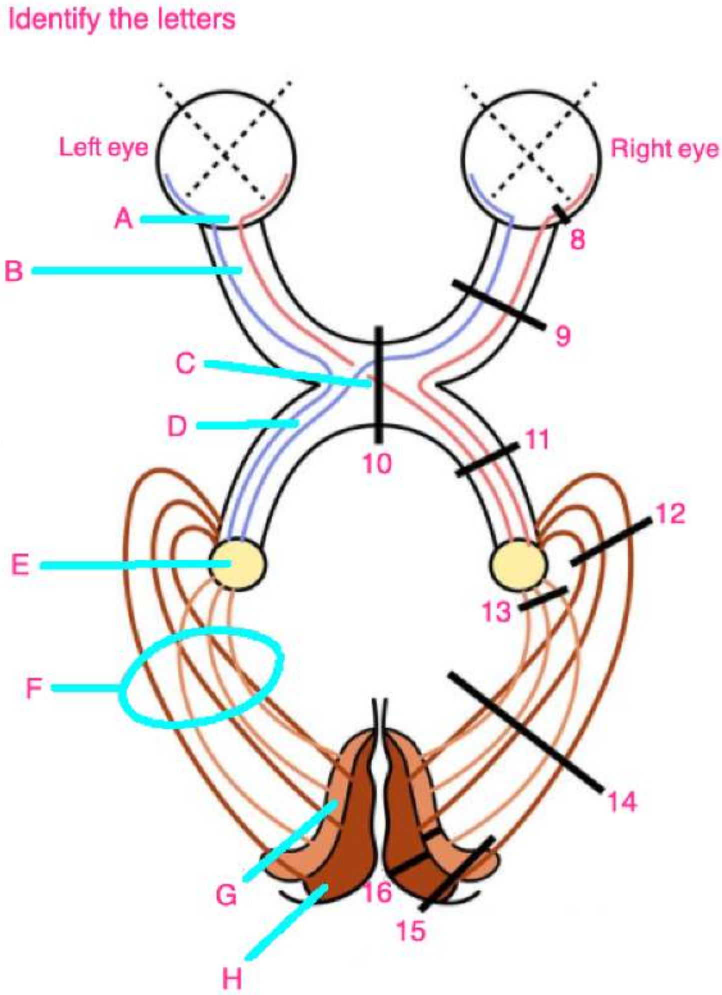
The most likely diagnosis is prolactinoma, which is the most common pituitary tumor in adults and can cause galactorrhea. Pituitary tumors impinge superiorly on the optic chiasm, leading to bitemporal hemianopsia. This is pass-level. Some of you doing this question thought this was too easy. That's fine. But the purpose of this PDF is to tell you what you need to know to pass USMLE, not to dance around obscure nonsense because I have nothing better to do than sit in a cafe right now here in Osaka writing up ophthal because I think it's fun.

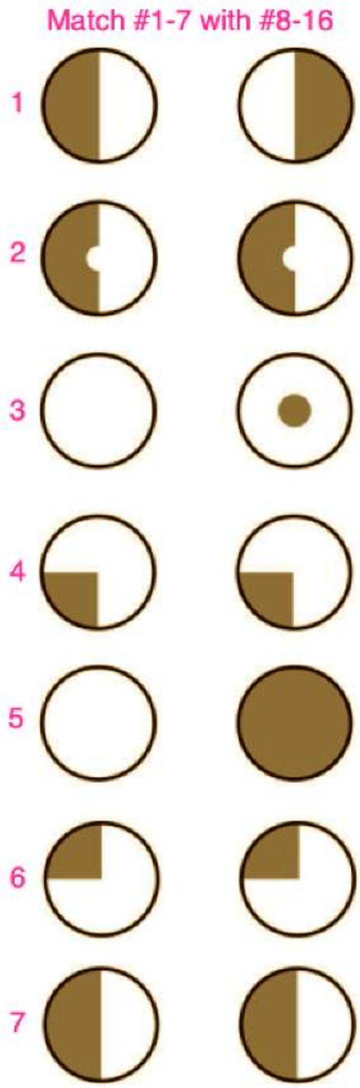
Memorize the below diagram.

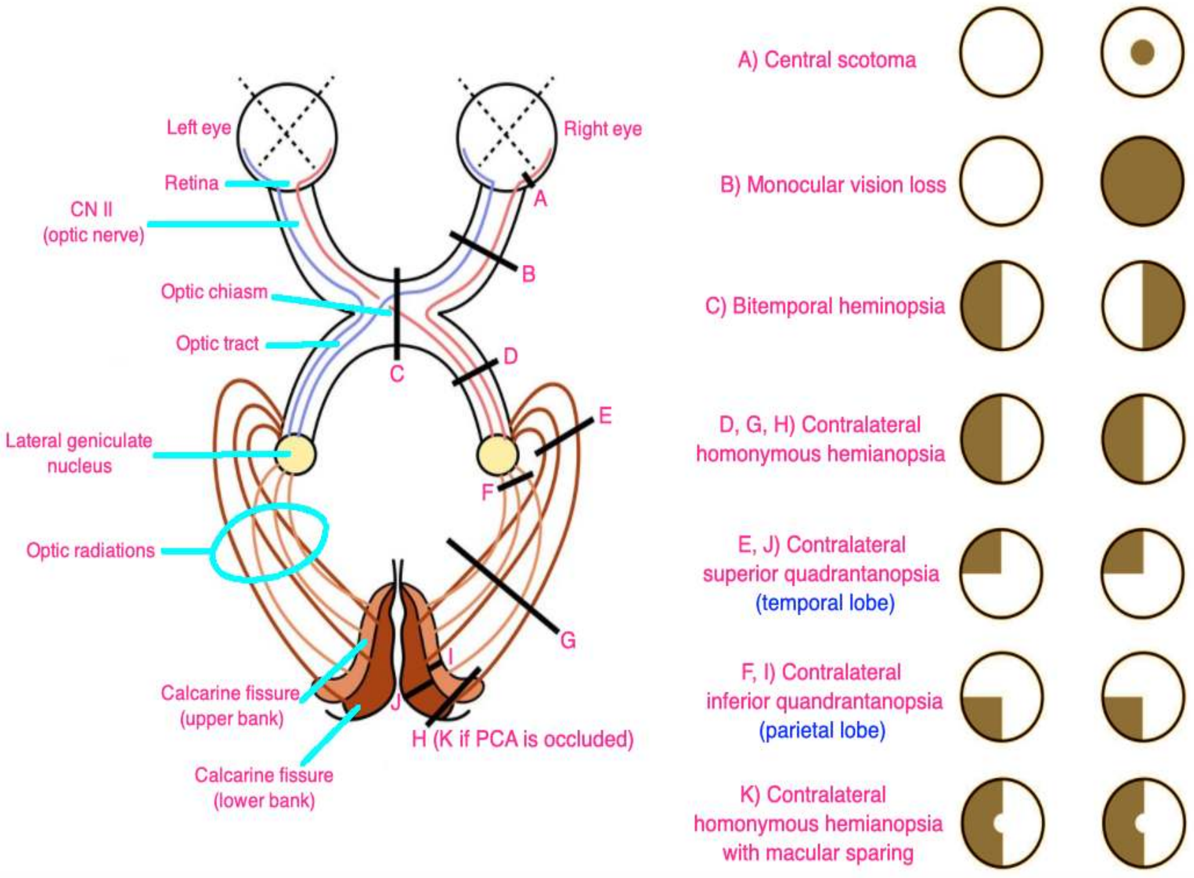


Did you memorize it? Because the next question will rely on you knowing it.

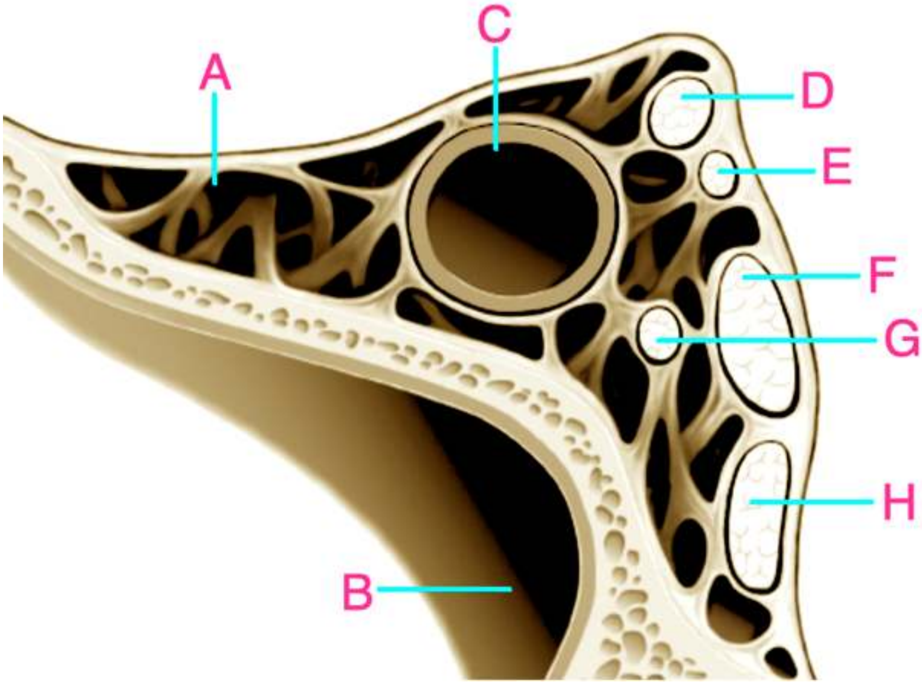
18) Annoying yes:






Left eye
Right eye
Retina
CN II
(optic nerve)
A
B
Optic chiasm
Optic tract
C
D
E
Lateral geniculate
nucleus
F
Optic radiations
G
Calcarine fissure
(upper bank)
J
I
H (K if PCA is occluded)
Calcarine fissure
(lower bank)
A) Central scotoma
B) Monocular vision loss
C) Bitemporal heminopsia
D, G, H) Contralateral
homonymous hemianopsia
E, J) Contralateral
superior quadrantanopsia
(temporal lobe)
F, I) Contralateral
inferior quandrantanopsia
(parietal lobe)
K) Contralateral
homonymous hemianopsia
with macular sparing

19) A 43-year-old woman presents to the ED with a severe headache, bilateral periorbital swelling, and impaired eye movements. Temperature is 103 F. She has had right cheek pain for the past three weeks and a runny nose. On examination, her right eye exhibits proptosis, ptosis, and limited extraocular movements, particularly affecting abduction. The left eye displays similar symptoms, but to a lesser degree. Her visual acuity is unaffected. Which of the following structures is the most likely explanation for her impaired abduction?



The answer is G.

The diagnosis is cavernous sinus thrombosis. This can occur due to sinusitis that ascends the valveless facial veins to the cavernous sinus, leading to clotting of blood. The classic triad for cavernous sinus thrombosis is headache, proptosis, and ophthalmoplegia. Ptosis can also be seen.

Cranial nerve VI can classically be affected, leading to impaired abduction.

A = cavernous sinus; B = sphenoid sinus; C = internal carotid artery; D = oculomotor nerve (CN III); E = trochlear nerve (CN IV); F = ophthalmic branch of trigeminal nerve (V1); G = abducens nerve (CN VI); H = maxillary branch of trigeminal nerve (V2).

20) An 8-year-old boy is brought to the physician by his mother for a 2-day history of swelling of the left eye. He recently had an upper respiratory tract infection. Temperature is 102 F. On examination, there is erythema and edema of the periorbital tissue, and difficulty opening the left eye due to the swelling. His eye movements are restricted and there is proptosis. Which of the following is the most likely diagnosis?

A) Chalazion
B) Hordeolum
C) Hyphema
D) Orbital cellulitis
E) Pre-septal cellulitis

The answer is D (orbital cellulitis).

This image is a spot-diagnosis. The child will look like he or she got hit in the eye with a softball. There can be history of recent upper respiratory tract infection, followed by ophthalmoplegia and/or proptosis. Even though cavernous sinus thrombosis can present with the latter two findings, the eye will not appear as such. Orbital cellulitis is a medical emergency and requires IV antibiotics.

Chalazion is a painless bump on the eyelid that is due to a blocked oil duct.

Hordeolum (sty) is a painful bump on the eyelid that is due to *S. aureus* infection.

Both chalazion and hordeolum are initially treated with warm compresses.

Hyphema is blood in the anterior chamber, usually following trauma to the eye.

Pre-septal cellulitis is less severe than orbital cellulitis and does not present with ophthalmoplegia and/or proptosis. Just remember basically: "Orbital cellulitis is the really bad one that looks like the kid got hit with a softball; pre-septal cellulitis is the one that's not as bad."

21) A 67-year-old man comes to the physician for a 4-month history of trouble swallowing and drooping of the left eyelid. Physical examination shows anisocoria, with the left pupil slightly smaller than the left. Further evaluation reveals decreased sweating on the left side of the face and neck. There is decreased pain and temperature sensation on the left side of the face and right side of the body. Which of the following is most likely responsible for this patient's condition?

A) Brachial plexus injury
B) Lateral medullary syndrome
C) Lateral pontine syndrome
D) Medial medullary syndrome
E) Syringomyelia

The answer is B (lateral medullary syndrome).

Lateral medullary syndrome (Wallenberg syndrome) is due to stroke of the ipsilateral posterior inferior cerebellar artery (PICA) or vertebral artery. USMLE wants you to know it is a HY cause of Horner syndrome.

In other words, Pancoast tumors of the lung (usually adenocarcinoma) impinging on the superior cervical ganglia are not the only cause of Horner syndrome for USMLE.

Horner syndrome = triad of ipsilateral miosis, partial ptosis, and anhidrosis.

For the stroke syndromes, just remember PICAchew (Pikachu) = PICA + dysphagia (can't swallow) + Horner.

Lateral pontine syndrome = AICA = anterior inferior cerebellar artery; AICA is in FACIAL spelled backwards → the stroke disorder that causes ipsilateral Bells palsy.

Medial medullary syndrome = anterior spinal artery stoke = ipsilateral tongue deviation (CN XII; hypoglossal palsy).

Brachial plexus injury for USMLE is either Erb-Duchenne palsy or Klumpke palsy. Both can be due to traumatic child birth in the neonate or in an older patient who's sustained an injury (e.g., grabbing onto a tree branch while falling).

Erb-Duchenne is upper brachial plexus injury (C5-6) and presents as arm in waiter-tip position.

Klumpke palsy is lower brachial plexus injury (C8-T1) and presents as a claw/clenched hand.

Syringomyelia is bilateral loss of pain and temperature due to a lesion at the anterior white commissure, which is the decussation point in the spinal cord for the spinothalamic tracts.

If you're reading this neuro stuff here and are confused as fuck, I recommend going to my HY Neuroanatomy PDF, which is Ace.

22) A 38-year-old African American woman comes to the physician for a 3-day history of a painful red eye and photophobia. Slit-lamp examination shows ciliary injection and small white cells floating in the anterior chamber. The patient also mentions experiencing shortness of breath and an ongoing dry cough. CXR shows bihilar lymphadenopathy. Which of the following is most likely to be seen in this patient?

A) Anterior uveitis; ↑ serum calcium; ↑ PTH; ↑ hydroxylase activity
B) Anterior uveitis; ↑ serum calcium; ↑ PTH; ↓ hydroxylase activity
C) Anterior uveitis; ↑ serum calcium; ↓ PTH; ↑ hydroxylase activity
D) Anterior uveitis; ↑ serum calcium; ↓ PTH; ↓ hydroxylase activity
E) Anterior uveitis; ↓ serum calcium; ↑ PTH; ↑ hydroxylase activity
F) Anterior uveitis; ↓ serum calcium; ↑ PTH; ↓ hydroxylase activity
G) Anterior uveitis; ↓ serum calcium; ↓ PTH; ↑ hydroxylase activity
H) Anterior uveitis; ↓ serum calcium; ↓ PTH; ↓ hydroxylase activity
I) Conjunctivitis; ↑ serum calcium; ↑ PTH; ↑ hydroxylase activity
J) Conjunctivitis; ↑ serum calcium; ↑ PTH; ↓ hydroxylase activity
K) Conjunctivitis; ↑ serum calcium; ↓ PTH; ↑ hydroxylase activity
L) Conjunctivitis; ↑ serum calcium; ↓ PTH; ↓ hydroxylase activity
M) Conjunctivitis; ↓ serum calcium; ↑ PTH; ↑ hydroxylase activity
N) Conjunctivitis; ↓ serum calcium; ↑ PTH; ↓ hydroxylase activity
O) Conjunctivitis; ↓ serum calcium; ↓ PTH; ↑ hydroxylase activity
P) Conjunctivitis; ↓ serum calcium; ↓ PTH; ↓ hydroxylase activity

The answer is C (Anterior uveitis; ↑ serum calcium; ↓ PTH; ↑ hydroxylase activity).

The diagnosis is sarcoidosis. Autoimmune diseases, not limited to sarcoidosis, can cause anterior uveitis. Ankylosing spondylitis and IBD are other HY conditions that do this.

Conjunctivitis is possible, but not the best answer here, because the stem says there is injection (redness) of the ciliary body, not the conjunctiva. The uvea is the ciliary body (muscle that controls the lens), the iris, and the choroid. The choroid is a vascular meshwork at the back of the eye and nourishes the retina. In anterior uveitis, there is inflammation of the ciliary body and iris.

The mechanism for sarcoidosis is non-caseating granulomas in the lungs composed of activated (epithelioid) macrophages (aka histiocytes) secreting 1α-hydroxylase. This converts inactive 25-D3 into active 1,25-D3. The latter goes to the small bowel and increases calcium absorption, causing hypercalcemia.

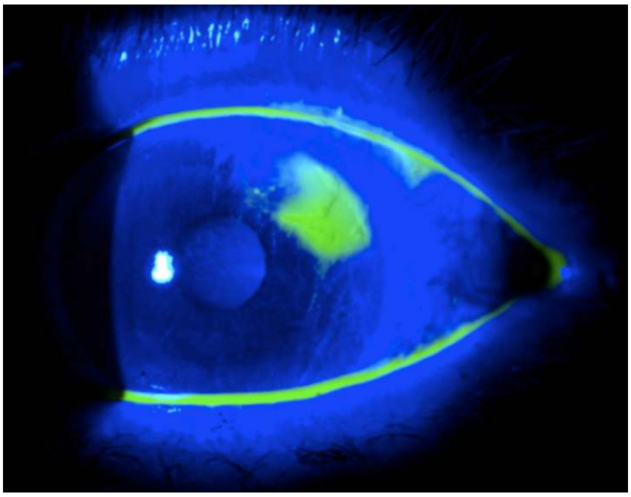
The hypercalcemia then suppresses endogenous PTH production due to negative feedback at the calcium-sensing receptors on the parathyroid glands.

It is HY you know (and I talk about this in the HY Arrows PDF) that 1,25-D3 is high and PTH is low in sarcoidosis.

23) A 46-year-old woman presents to emergency for a 1-hour history of intensely scratchy and teary eye. She wears contacts. There is no past medical history. Which of the following is the next best step in management?

A) Administer cycloplegic drops and schedule a follow-up in 24 hours
B) Antibiotic ointment
C) Fluorescein instillation into the eye
D) Fundoscopy
E) Oral antibiotic and a topical anesthetic

The answer is C (fluorescein instillation into the eye).



The diagnosis is corneal abrasion. USMLE loves this blue-green picture. Anything blue is normal. Anything green is abnormal.

Fluorescein is an orange dye that when dripped into the eye appears blue-green under violet light.

Cycloplegics paralyze the ciliary muscle and dilate the eye. These are often used to conduct slit-lamp examinations.

Fundoscopy visualizes the retina. This is not the most urgent next best step.

Once diagnosis of corneal abrasion is made, the patient may receive topical antibiotics (i.e., topical fluoroquinolone). But the diagnosis via fluorescein instillation comes first.

24) A 28-year-old woman comes to the physician for a follow-up appointment. Physical examination shows a notch-like defect in the right iris. Which of the following is most likely to be seen in this patient?

A) Atrial septal defect
B) Epicanthal folds
C) Eustachian tube atresia
D) Hypocalciuric hypecalcemia
E) Hypothyroidism

The answer is A (atrial septal defect).

The image is showing coloboma of the eye, which means hole in the eye. The diagnosis is CHARGE syndrome, which stands for Coloboma of the eye, Heart defects, Atresia of the choanae, Retardation of growth and development, Genitourinary anomalies, Ear defects.

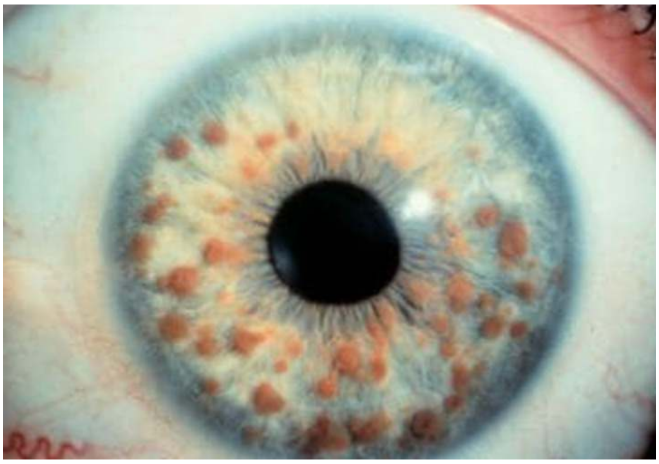
This is a 2CK Peds diagnosis. If you are studying for Step 1, relax. It's okay to learn new stuff.

Don't confuse coloboma with aniridia, which means absent iris. This is seen in WAGR, which is Wilms tumor, Aniridia, Genitourinary anomalies, Retardation.

Epicanthal folds (skin covering medial part of eye) and Eustachian tube atresia are classic for, but not limited to, Down syndrome.

Hypocalciuric hypercalcemia is a benign familial condition that has no relation here. In hyperparathyroidism, urinary calcium is high, not low, even though PTH pulls calcium out of the urine. This is because hypercalcemia leads to high filtration of calcium into the urine, so even though PTH pulls it out of the urine, the urinary calcium is still high. In familial hypocalciuric hypercalcemia, in contrast, you guessed it, there is low urinary calcium despite hypercalcemia. It's nothing to obsess about and ultra LY. Just threw in here as distractor.

25) A 25-year-old man comes to the physician for a routine health maintenance examination. Physical examination shows axillary and groin freckling. He has hamartomatous lesions within his irides. There are also scattered hyperpigmented macules on his trunk and limbs. The probability one of his children will inherit this condition is most close to which of the following?



A) 0% due to X-linked recessive inheritance
B) 25%
C) 50%
D) 100%
E) Only female offspring will be affected
F) Undeterminable unless carrier state of female is known

The answer is C (50%).

The diagnosis is neurofibromatosis type I (NF1). It is an autosomal dominant condition on chromosome 17, which means patients almost always only have one mutation. There is a 50% chance he will pass this to both male and female children.

The image shows Lisch nodules, which are iris hamartomas. USMLE doesn't expect you to identify this image where you say, "Oh obviously those are Lisch nodules." But they want you to know that iris hamartomas are seen in NF1.

NF1 can present with axillary and groin freckling, café au lait spots (hyperpigmented macules), **optic glioma (tumor of CN II),** neurofibromas (nodules under the skin), pheochromocytoma, and weird brain cancers such as ependymoma and oligodendroglioma.

26) A 7-day-old neonate is brought to the pediatrician by his mother for profuse yellow discharge from his left eye since birth. Examination shows chemosis and palpable preauricular lymph nodes. He is afebrile. Which of the following is the most likely explanation for these findings?

A) Allergic conjunctivitis
B) Chemical conjunctivitis
C) Chlamydial conjunctivitis
D) Gonococcal conjunctivitis
E) Viral conjunctivitis

The answer is D (gonococcal conjunctivitis).

Ophthalmia neonatorum (neonatal conjunctivitis) in the first week of life accompanied by discharge is classically gonococcal.

If the USMLE wants chlamydia, they will give a more chronic course beyond the first week that **leads to pneumonia** (as discussed earlier) due to drainage through the nasolacrimal duct. In other words, basically any chlamydial conjunctivitis Q will mention something about crackles in the lung fields / a pneumonia, since this is so HY.

Prophylaxis for chlamydial conjunctivitis is treating the mom while pregnant if she has an infection; treatment in the neonate who already has the conjunctivitis is **oral** erythromycin.

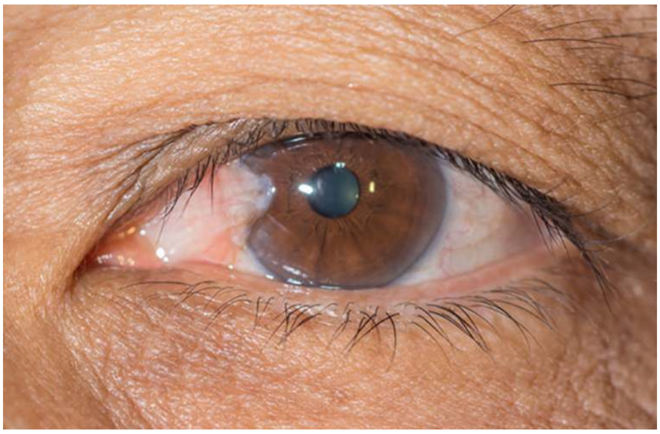
Prophylaxis for gonococcal is erythromycin ointment; treatment is IM 3rd-gen cephalosporin (usually cefotaxime in peds).

Allergic conjunctivitis will present as cobblestoning of the nasal mucosa in a child or adult. Patients can often have atopy (asthma, eczema, hay fever).

Chemical conjunctivitis is classically due to silver nitrate eyedrops which used to be given to neonates for gonococcal prophylaxis, however they have been replaced by erythromycin ointment.

Viral conjunctivitis will present as teary eye that is either unilateral or bilateral, and either itchy or non-itchy. I've seen variable presentation on USMLE. You just need to know this is the most likely diagnosis in both school-age kids and adults, and is caused by **adenovirus,** which is a DNA virus. Treatment is supportive with saline rinse.

27) A 59-year-old man comes to the physician for a health maintenance examination. He works in construction and has no past medical history. Which of the following is the most likely diagnosis?



A) Apigmented melanoma
B) Chalazion
C) Hordeolum
D) Pinguecula
E) Pterygium

The answer is E (pterygium).

Pterygium is a non-cancerous overgrowth of non-keratinized stratified squamous epithelium growing from the conjunctiva. It crosses over the cornea (clear part of the eye covering the iris and pupil). History of sun exposure is the major risk factor.

Pinguecula is a yellow-white benign lesion on the conjunctiva composed of protein and fat. It does not cross over the cornea. Sunlight is the major risk factor.

Apigmented conjunctival melanoma can occur in patients with sunlight exposure, but this image showing a fleshy growth of tissue over the cornea is a spot-diagnosis for pterygium.

Chalazion is a painless bump on the eyelid that is due to a blocked oil duct.

Hordeolum (sty) is a painful bump on the eyelid that is due to *S. aureus* infection.

Both chalazion and hordeolum are initially treated with warm compresses.

28) A 31-year-old man presents to the ED with lancinating periorbital pain behind his left eye. He describes it as a "stabbing" pain that wakes him from sleep. He has had near-daily episodes for the past few weeks, which typically last 15-30 minutes. They go away when he paces around his room. On physical examination, there is ptosis and miosis of the left eye. Which of the following is the most likely diagnosis?

A) Cluster headache
B) Herpetic neuralgia
C) Horner syndrome
D) Temporal arteritis
E) Trigeminal neuralgia

The answer is A (cluster headache).

Cluster headache will present on USMLE as a male 20s-40s with severe lancinating pain behind the eye / in the temporal region that usually wakes him from sleep. Episodes tend to last around 15-30 minutes. They can be associated with rhinorrhea, ptosis, and miosis. Treatment is with hyperbaric oxygen. Prophylaxis is verapamil.

Herpetic (or post-herpetic) neuralgia is neuropathic pain from Herpesviridae (i.e., HSV1/2 and VZV) that can occur before, during, and after vesicular eruption. Gabapentin is a classic agent that can mitigate herpetic pain. In the case of VZV, by the way, it should be noted this refers classically to shingles.

Horner syndrome is the triad of ipsilateral partial ptosis, miosis, and anhidrosis seen with Pancoast tumors of the lung impinging on the superior cervical ganglia, as well as in lateral medullary (Wallenberg) syndrome.

Temporal arteritis presents as severe unilateral (although one 2CK NBME Q gives it bilateral) temporal headache in a patient over 50. It may be associated with jaw claudication (pain in the jaw with chewing) and polymyalgia rheumatica (pain and stiffness of proximal muscles, without an increase in creatine kinase or weakness on physical exam; if either of the latter two is present, diagnosis is polymyositis instead). Give IV steroids (methylprednisolone) prior to biopsy to prevent blindness.

29) A 42-year-old man with schizophrenia is admitted to the psychiatric unit for acute psychosis and is initiated on haloperidol and benztropine. Four days after starting the medication, he develops confusion, agitation, and visual hallucinations. Pupils are 4mm. Temperature is 100.3 F. What is the most likely cause of the patient's presentation?

A) Anti-cholinergic delirium
B) Delirium tremens
C) Drug allergy
D) Malignant hyperthermia
E) Neuroleptic malignant syndrome

The answer is A (anti-cholinergic delirium).

I've seen NBME give 4mm pupils as large. Students have asked before what is considered "large" for pupils.

Anti-psychotics such as haloperidol can have anti-cholinergic side-effects. In addition, the patient is on benztropine, which is an anti-cholinergic (muscarinic receptor antagonist) used to treat acute dystonia due to anti-psychotics. In other words, the fact that benztropine is mentioned here should scream anti-cholinergic effects as you're reading the stem.

Cholinergic effects = DUMBBELSS = Diarrhea, Urination, Miosis, Bradycardia, Bronchoconstriction, Excitation (neuromuscular), Lacrimation, Salivation, Sweating.

Anti-cholinergic side-effects are the "opposite of DUMBBELSS" = Constipation, Urinary retention, Mydriasis, Tachycardia, Bronchodilation, Flaccidity, Dry eyes, Dry mouth, Anhydrosis.

Patients with anti-cholinergic side-effects can appear hot and dry (due to inability to sweat). USMLE also loves giving a "suprapubic mass palpated" in order to denote urinary retention.

The temperature is not high enough for neuroleptic malignant syndrome (NMS). The Q will give a temperature bare minimum 102-103 F. This is due to ryanodine channel getting stuck open, which allows calcium to move from the sarcoplasmic reticulum into the cytosol. The cell then needs to utilize ATP to get the calcium back into the sarcoplasmic reticulum, and this generates heat. Dantrolene inhibits ryanodine channel.

Malignant hyperthermia is the same mechanism as NMS but is caused classically by succinylcholine during general anesthesia. Succinylcholine is a depolarizing nicotinic receptor antagonist that paralyzes muscles.

Drug allergy can present on USMLE as urticaria (hives), anaphylaxis, maculopapular rash, Stevens-Johnson syndrome, toxic epidermal necrolysis, tubulointerstitial nephritis, or high eosinophils on bloods.

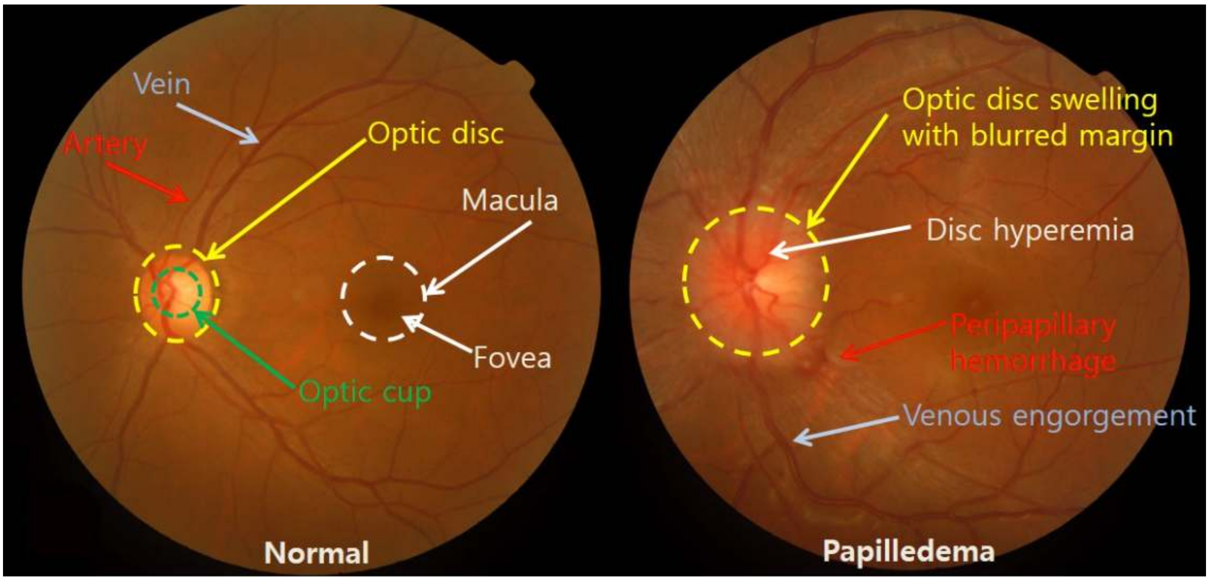
30) A 29-year-old woman presents to the physician with severe headache, nausea, and vomiting for the past 5 days. BMI is 33. Blood pressure is 140/90. Heart rate is 55. Fundoscopy shows papilledema and absent venous pulsations. A non-contrast head CT scan shows no abnormalities. Lumbar puncture shows an elevated opening pressure. Which of the following is the most likely explanation for these findings?

A) Brain tumor
B) Cerebral venous sinus thrombosis
C) Idiopathic intracranial hypertension
D) Medication adverse effect
E) Normal pressure hydrocephalus

The answer is C (idiopathic intracranial hypertension; IIH).

It is the answer on USMLE for an overweight woman who shows signs of increased intracranial pressure.

Papilledema and absent venous pulsations are buzzy descriptors for increased intracranial pressure. Papilledema means blurring of the optic disc margins due to inflammation of the optic nerve head from the increased intracranial pressure. The USMLE does not expect you to know the fundoscopy for this. They just want you to know the word "papilledema" = increased intracranial pressure, that's all.



Brain tumors on USMLE will often present with morning vomiting and seizure. This answer is wrong because the CT of the head showed no abnormalities.

Cerebral venous sinus thrombosis is a nonsense answer choice that doesn't show up on USMLE. Just decided to be an unremitting asshole and throw in some weird distractor.

Hard for the answer to be medication adverse effect when the vignette mentions zero about medications. Vitamin A, danazol, and OCPs can cause pseudotumor cerebri, which is just another way of saying increased intracranial pressure that is either idiopathic (as with this question) or due to a medication.

Normal pressure hydrocephalus presents as "wet, wobbly, wacky" +/- Parkinsonism in a patient over 50. The lateral ventricles are enlarged on CT of the head.

31) A 51-year-old woman comes to the physician for a 2-month history of eye discomfort that she describes as dry, gritty, and itchy. She also reports a burning sensation and frequent episodes of blurred vision, particularly when reading or using a computer for extended periods. On examination, there is bilateral conjunctival hyperemia and minimal tear film meniscus. Fundoscopy does not show debris in the anterior chamber or erythema of the ciliary body. A Schirmer test reveals decreased tear production. Laboratory studies show positive titers for anti-nuclear antibody. Which of the following is the most likely explanation for this patient's findings?

A) Allergic conjunctivitis
B) Anterior uveitis
C) Fuchs dystrophy
D) Keratoconjunctivitis sicca
E) Ophthalmitis medicamentosa

The answer is D (keratoconjunctivitis sicca).

Keratoconjunctivitis sicca is dry eye (xerophthalmia) caused by decreased tear production. It is classically autoimmune, either as a stand-alone condition or as part of Sjögren syndrome. The latter presents additionally with dry mouth (xerostomia) and systemic findings such as arthritis.

For allergic conjunctivitis, the USMLE will mention “cobblestoning” of the tarsal conjunctiva. Similarly, if the patient has allergic rhinitis, the vignette can say cobblestoning of the nasal mucosa. This is an extremely HY buzz word for USMLE.

Anterior uveitis is seen in autoimmune diseases, and hence +ANA could be seen in the vignette, but the stem would not overtly tell you there are negative cells in the anterior chamber or a non-erythematous ciliary body. The anterior uvea is the iris and ciliary body.

Fuchs dystrophy you do not need to know for USMLE. This is clouding of the cornea due to corneal endothelial cell dysfunction and an inability to regulate its level of hydration and transparency.

Ophthalmitis medicamentosa (aka conjunctivitis medicamentosa) is eye inflammation due to topical medication use, such as eye drops. Chemical conjunctivitis, is a broader term that can refer to irritants, not limited to medicine, causing irritation of the eye (e.g., fumes).

32) A 5-year-old girl is brought to the physician by her mother due to concern about her eyes. She has difficulty with depth perception and occasionally has double vision. Examination shows the right eye turns inward when focusing on near objects. Which of the following is the most likely explanation for this patient's findings?

A) Amblyopia
B) Astigmatism
C) Hyperopia
D) Myopia
E) Strabismus

The answer is E (strabismus).

Strabismus means “lazy eye,” where there is weakening of muscles in one eye, leading to misalignment. This condition can lead to amblyopia, which is reduced visual acuity in the weak eye due to the brain favoring the stronger eye during development. Strabismus, however, is not reduced vision in the weak eye. It precedes amblyopia if not treated with eye exercises and/or sometimes penalization of (patching) the strong eye.

Astigmatism is reduced visual acuity due to irregular contour of the cornea, where the shape, rather than being round like a sphere, could be described as having valleys or mountains.

Hyperopia is far-sightedness, which is the ability to see far objects better than near objects due to an eyeball or cornea that is too short anteroposteriorly. This is not the same as presbyopia, which is reduced ability to see near objects due to stiffening of the lens and ciliary body with age.

Myopia is near-sightedness, which is the ability to see near objects better than far objects due to an eyeball or cornea that is too long anteroposteriorly. Severe myopia is a risk factor for retinal detachment, as discussed earlier.

33) A 60-year-old man is brought to emergency by his wife for a severe headache accompanied by vomiting. Neurological examination shows a dilated right pupil that is non-reactive to light directly and consensually. The left eye is normal. His neck is stiff. Vitals are normal. Which of the following is the most likely location of the patient's pathology?

A) Anterior cerebral artery
B) Anterior communicating artery
C) Bridging vein
D) Middle meningeal artery
E) Posterior communicating artery

The answer is E (posterior communicating artery).

Severe headache can suggest a rupturing anterior (ACOM) or posterior communicating artery (PCOM) aneurysm, as either can lead to subarachnoid hemorrhage. However, PCOM aneurysm (ruptured or not) can lead to an ipsilateral blown pupil (i.e., ipsilateral mydriasis) due to impingement on CN III, as I mentioned earlier in this PDF.

The stiff neck (aka meningismus; meningism) is not meningitis. It is HY you know that subarachnoid hemorrhage, with blood irritating the meninges, is another cause of stiff neck.

The anterior cerebral artery (ACA) is not a typical location for aneurysm. ACA strokes, however, will cause motor or sensory dysfunction of the contralateral leg.

The middle cerebral artery (ACA) is not a typical location for aneurysm. MCA strokes, however, will cause motor or sensory dysfunction of the contralateral arm and face. If dominant-side (usually the left brain), it can also cause Wernicke or Broca encephalopathy. If non-dominant (usually the right brain), it can lead to hemispatial neglect and inability to draw a clockface properly.

Bridging vein rupture leads to subdural hematoma. This is a crescent moon-shaped bleed seen classically in elderly with dementia who've had falls, alcoholics, and deceleration injury (i.e., car accidents; shaken baby syndrome). There is no lucid interval.

Middle meningeal artery rupture causes epidural hematoma. This is a biconvex lens-shaped bleed that occurs with hitting one's head, classically in the temporal area. There is a lucid interval, which means initially losing consciousness for a few minutes, then awakening with clarity for minutes to hours (i.e., lucid interval, where the patient says "Don't call an ambulance I'm fine."), followed by going home to sleep and dying. This is usually due to brainstem herniation from increased pressure from the bleed.

34) A 3-year-old girl is evaluated for conductive hearing loss due to Eustachian tube dysfunction and middle ear fluid collections. Physical examination shows slanted palpebral fissures and single palmar creases. One of the patient's eyes is shown both under visible and near-infrared wavelengths. The mother is 47 years old. Which of the following is the most likely explanation for these findings?

A) Bitot spots
B) Brushfield spots
C) Corneal arcus
D) Hemangioblastomas
E) Lisch nodules

The correct answer is B (Brushfield spots).

The diagnosis is Down syndrome (trisomy 21). The images show Brushfield spots, which are connective tissue deposits seen within the irides of some patients with Down syndrome.

Eustachian tube dysfunction and middle ear fluid collection ("glue ear") leading to conductive hearing loss is common in Down syndrome. It should be noted that single palmar crease, although frequently mentioned in Down syndrome questions, is non-specific. There is a 2CK psych form that has single palmar crease mentioned in a fetal alcohol syndrome question.

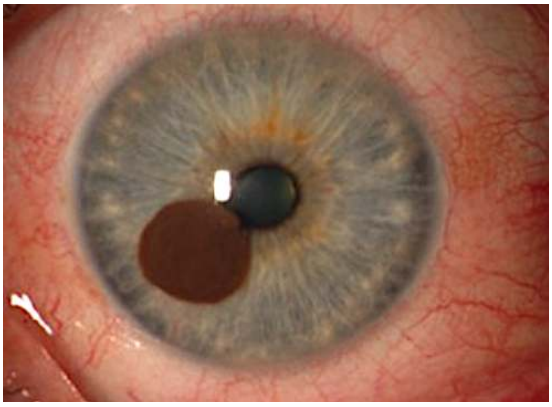
Bitot spots are conjunctival lesions that appear on the eye in vitamin A deficiency.

Corneal arcus is a whitish ring of lipid deposits that occurs in the eye in some patients due to normal aging. It can also be a sign of hyperlipidemia.

Retinal and cerebellar hemangioblastomas can be seen in von Hippel Lindau (VHL) in conjunction with renal cell carcinoma. Pancreatic cysts can also be seen.

Lisch nodules are iris hamartomas seen in NF1, as mentioned earlier.

35) A 74-year-old woman in Florida presents to the ophthalmologist for a slowly growing on the eye. The patient has no past medical history. Which of the following is the most likely diagnosis?



A) Coloboma
B) Hyphema
C) Melanoma
D) Pinguecula
E) Pterygium

The answer is C (melanoma).

Melanomas can grow from the conjunctiva and cornea. Just be aware they exist.

Coloboma is means a hole in the eye. It presents as a notch-like defect in the iris and is seen in CHARGE syndrome: Coloboma of the eye, Heart defects, Atresia of the choanae, Retardation of growth and development, Genitourinary anomalies, Ear defects.

Hyphema is blood in the anterior chamber.

Pinguecula is a yellow-white benign lesion on the conjunctiva composed of protein and fat. It does not cross over the cornea. Sunlight is the major risk factor.

Pterygium is a non-cancerous overgrowth of non-keratinized stratified squamous epithelium growing from the conjunctiva. It crosses over the cornea (clear part of the eye covering the iris and pupil). History of sun exposure is the major risk factor.

36) An 8-year-old boy is brought to the physician for a 1-month history of worsening vison and hearing. Physical examination shows diffuse hypotonia. Laboratory studies show a bicarbonate of 21 (NR 22-28 mEq/L). His mother has poor hearing. He has an older brother with normal vision but poor hearing. Which of the following is the most likely explanation for this patient's findings?

A) Friedreich ataxia
B) Fuchs dystrophy
C) Leber hereditary optic neuropathy
D) Myotonic dystrophy
E) Niemann-Pick disease
F) Tay-Sachs disease

The answer is C (Leber hereditary optic neuropathy).

The USMLE doesn't expect you to know the hyper-specifics of this condition. They just want you to know it is one of the mitochondrial disorders, which present as triad of 1) eye/ear problems, 2) hypotonia, and 3) lactic acidosis (low bicarb; metabolic acidosis). Other mitochondrial disorder names you could be aware of are MELAS (Mitochondrial Encephalopathy, Lactic Acidosis, and Stroke-like episodes) and MERRF (Myoclonic epilepsy with ragged red fibers).

Friedreich ataxia is a trinucleotide repeat (TNR) disorder (GAA) that presents as ataxia (you guessed it), cardiomyopathy, pes cavus (high-arched feet), and scoliosis.

Fuchs dystrophy you do not need to know for USMLE. This is clouding of the cornea due to corneal endothelial cell dysfunction and an inability to regulate its level of hydration and transparency.

Myotonic dystrophy is another TNR disorder (CTG) that presents with an inability to relax muscles (myotonia) after actions such as grabbing a doorknob, shaking a person's hand, or using a golf club. Cataracts can also be seen.

Niemann-Pick disease (NPD) is a lysosomal storage disease caused by a deficiency of sphingomyelinase, leading to a buildup of sphingomyelin. NPD can present as vision loss in a young child (causing cherry-red spot on the macula) and neurodegeneration. There is also hepatosplenomegaly and foam cells (lipid-laden macrophages).

Tay-Sachs disease is a lysosomal storage disease caused by a deficiency of hexosaminidase A, leading to a buildup of GM2 ganglioside. Similar to NPD, Tay-Sachs can present with vision loss in a young child and cherry-red spot on the macula) and neurodegeneration, however there is no hepatosplenomegaly. Hyperacusis is a common finding in Tay-Sachs.

37) A 29-year-old male comes to the physician for a 1-month history of worsening hearing and red urine. He has no past medical history. Family history is unremarkable. He is 6’3”. Which of the following is the most likely diagnosis?

A) Alport syndrome
B) Behcet disease
C) Goodpasture syndrome
D) Homocystinuria
E) Marfan syndrome

The answer is A (Alport syndrome).

Alport syndrome is an X-linked recessive mutation in type IV collagen. It presents as a male who has hematuria (red urine) and eye and/or ear problem (i.e., diminished vision or hearing).

Don’t confuse Alport with Goodpasture syndrome, which is due to antibodies against type IV collagen (aka anti-glomerular basement membrane; anti-GBM) antibodies. The presentation is hematuria and hemoptysis in a male 20s-40s.

Behcet disease is an autoimmune disorder causing oral and genital ulcers. All you need to know is that combo = Behcet on USMLE.

Homocystinuria will present as a school-age kid with Marfanoid (i.e., tall, lanky) body habitus + lens dislocation + Hx of thrombosis. LY for USMLE, but you could be aware that it is a differential of Marfan syndrome.

38) A neonate born at term is found to have chorioretinitis, hydrocephalus, and intracranial calcifications. The mother's pregnancy was unremarkable. Which of the following is the most likely neonatal diagnosis?

A) Congenital CMV
B) Congenital herpes
C) Congenital rubella
D) Congenital syphilis
E) Congenital toxoplasmosis

The answer is E (congenital toxoplasmosis).

You need to memorize for USMLE that the strict triad of 1) chorioretinitis, 2) hydrocephalus, and 3) intracranial calcifications is congenital toxoplasmosis. The vignette need not say anything about interaction with cats or consumption of pork. Toxoplasmosis in an older child or adult will present with ring-enhancing lesion(s) on CT of the head.

Congenital CMV can notably present with deafness. It is a diagnosis of exclusion on USMLE, meaning that you eliminate the other answer choices to get there. For example, they can say a neonate has deafness + intracranial calcifications + hepatomegaly + blue berry muffin rash, and you say, "Well calcifications are seen in toxo, but the classic toxo triad isn't there, so it can't be toxo." The other findings (i.e., hepatomegaly + blue berry muffin rash) are non-specific and can be seen in other conditions such as congenital rubella and syphilis.

Congenital rubella syndrome presents with patent ductus arteriosus (continuous machine-like murmur). There can also be cataracts and deafness.

Congenital syphilis presents with saddle nose, bone deformities (i.e., saber shins), and tooth anomalies (i.e., mulberry molars). Similar to congenital rubella, there can also be cataracts and deafness.

39) A 26-year old man comes to the physician for a 3-month history of wobbly gait. Physical examination shows a negative Romberg test. Fundoscopic examination shows a retinal lesion. His father died from renal cell carcinoma. Which of the following is the most likely diagnosis?

A) Neurofibromatosis type I (NF1)
B) Neurofibromatosis type II (NF2)
C) Tuberous sclerosis (TSC)
D) von Hippel Lindau (VHL)
E) Sturge-Weber syndrome

The answer is D (VHL).

All of the conditions listed in the above answer choices are collectively known as the phakomatoses (weird word that means neurocutaneous disorders).

VHL presents as cerebellar and retinal hemangioblastomas, renal cell carcinoma, and sometimes pancreatic cysts. The patient's ataxia is presumably from a cerebellar lesion. The negative Romberg test means proprioception is intact and that the etiology for the ataxia is not the dorsal columns. In other words, if the patient falls over with the Romberg test (i.e., positive test), the etiology is the dorsal columns on USMLE (e.g., with tabes dorsalis in neurosyphilis or B12 deficiency).

NF1 presents with axillary and groin freckling, café au lait spots (hyperpigmented macules), **optic glioma (tumor of CN II),** neurofibromas (nodules under the skin), pheochromocytoma, and weird brain cancers such as ependymoma and oligodendroglioma.

NF2 presents with acoustic schwannoma (need not be bilateral) and sometimes cataracts or meningioma.

TSC presents adenoma sebaceum (angiofibromas; i.e., peach-colored papules on the face), shagreen patches (hyperpigmented velvety skin lesions), ashleaf spots (hypopigmented macules), cardiac rhabdomyoma, renal angiomyolipoma, and subungual fibromas (nailbed tumors).

The above disorders are exceedingly HY for USMLE. Know these ice fucking cold.

40) A neonatal male is evaluated by an ophthalmologist. An image is shown. Which of the following is most likely to develop in this patient?

A) Cardiac defects
B) Choanal atresia
C) Coloboma of the eye
D) Ear anomalies
E) Wilms tumor

The answer is E (Wilms tumor).

The image is showing aniridia, which is part of WAGR complex (Wilms tumor, Aniridia, Genitourinary anomalies, Retardation).

Even if you have no idea what you are looking at in the picture, the other answers are all seen in CHARGE syndrome, which is how you know they're all wrong. Wilms tumor is the odd one out.

CHARGE syndrome = Coloboma of the eye, Heart defects, Atresia of the choanae, Retardation of growth and development, Genitourinary anomalies, Ear defects.

41) A 22-year-old man is brought to the ED following a skiing accident in which he hit a large tree branch. He presents with periorbital swelling, ecchymosis, and severe pain in his right eye. Examination shows restriction of upward and downward gaze in the right eye, along with diplopia. Pupillary reaction is normal direct and consensual, and visual acuity is normal. A CT scan reveals a fracture of the orbital floor. Which of the following structures is most likely to be injured in this patient's orbital fracture?

A) Inferior rectus and inferior oblique muscles
B) Lateral rectus muscle
C) Medial rectus muscle
D) Superior oblique muscle
E) Trochlear nerve

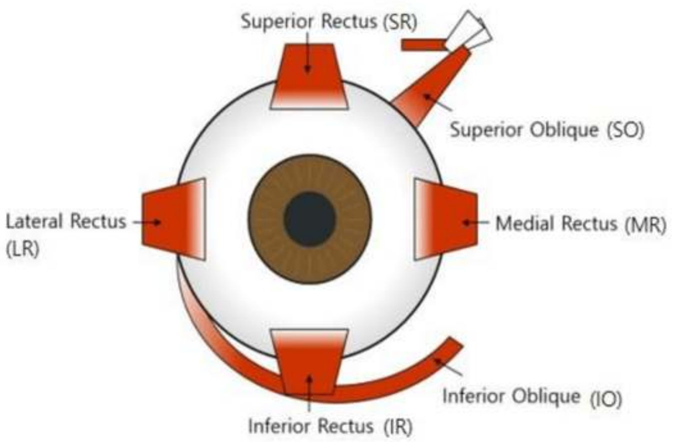
The answer is A (Inferior rectus and inferior oblique muscles).

It is assessed on NBME that orbital floor fractures can lead to damage to these two muscles due to the proximity of these muscles to the orbital floor. They are responsible for controlling downward (inferior rectus) and upward gaze (inferior oblique). Injury can result in limited vertical eye movement, causing diplopia.

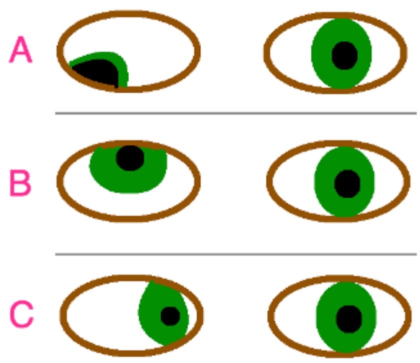
If the lateral rectus is fucked up, we'd have a medially-directed eye (esotropia; i.e., an eye that looks inward), similar to an abducens nerve palsy.

Medial rectus damage would cause exotropia, or a laterally-directed eye, due to inability to adduct.

Superior oblique muscle damage would cause a reduced ability to look downward. However, it is located more toward the upper aspect of the orbit, making it less likely to be injured in orbital floor fracture. Inability to perform downward gaze in the setting of orbital floor fracture suggests inferior rectus injury in this regard.



Trochlear nerve palsy results in an eye with hypertropia (upward-directed; as with B below).



A = lesion of CN III (oculomotor) → presents as down and out; B = CN IV lesion (trochlear) → presents as upward gaze + patient will tilt head toward unaffected side; C = CN VI (abducens) → medially directed eye.

Eye that is outward-directed = exotropia.
Eye that is inward-directed = esotropia.
Eye that is upward-directed = hypertropia.
Eye that is downward-directed = hypotropia.

USMLE won't test those vocabulary words, but I'm just clarifying/summarizing for the sake of this explanation.

42) A 38-year-old woman comes to the physician for a 6-month history of worsening diplopia and eyelid drooping toward the end of the day. She also reports occasional difficulty with swallowing food. Physical examination shows difficulty rising from seated position without using the arms. Laboratory studies show normal creatine kinase. She is a non-smoker. Chest x-ray shows a mediastinal mass. Which of the following is the most likely explanation for these findings?

A) Decreased presynaptic release of acetylcholine
B) Decreased presynaptic release of glycine
C) Inflammatory infiltrate within muscle fiber
D) Presence of antibodies against presynaptic ligand-gated calcium channels
E) Presence of antibodies against presynaptic voltage-gated calcium channels
F) Presence of antibodies against postsynaptic muscarinic acetylcholine receptors
G) Presence of antibodies against postsynaptic nicotinic acetylcholine receptors

The answer is G (Presence of antibodies against postsynaptic nicotinic acetylcholine receptors).

The diagnosis is myasthenia gravis, which can be a paraneoplastic syndrome of thymoma (although it of course can be idiopathic without thymoma). There is an NBME exam that assesses muscarinic versus nicotinic, and you need to know more specifically that nicotinic ACh receptors are the answer.

The triad of diplopia, ptosis, and dysphagia in an office worker in her 30s-50s is hyper-classic of MG.

Patients have muscle weakness that gets worse with activity. They might say the patient is unable to perform upward gaze for 60 seconds.

Presence of antibodies against presynaptic voltage-gated calcium channels refers to Lambert-Eaton syndrome, often a paraneoplastic of small cell bronchogenic carcinoma, which secretes the antibodies. LES presents as proximal muscle weakness that improves with activity. For example, the Q might say the patient tries a few times to get up from a chair, but after doing this, can then stand up easily. They might also say the patient is able to perform upward gaze for 60 seconds without a problem.

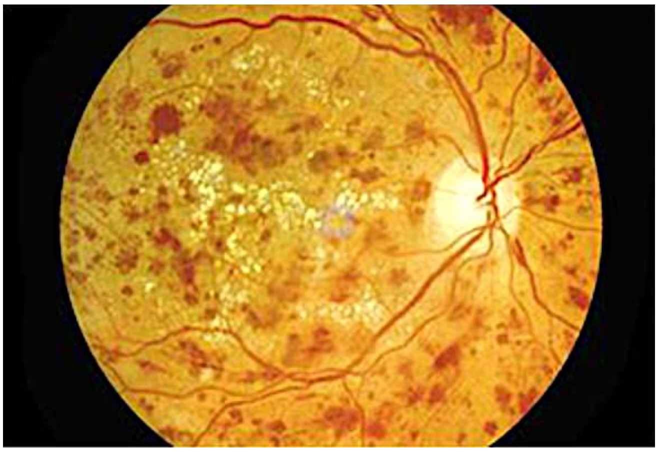
Decreased presynaptic release of acetylcholine is botulism (due to *Clostridium botulinum*). This will present as muscle flaccidity, constipation, and cranial nerve palsy. In children under the age of 1, this can be due to consumption of spores in honey. In older children and adults, this can be due to consumption of pre-formed toxin in contaminated canned good. Acetylcholine is stimulatory of muscles, so reduced ACh means flaccidity.

Decreased presynaptic release of glycine refers to tetanus (due to *Clostridium tetani*). Glycine is an inhibitory neurotransmitter, so decreased release means overstimulation of muscle.

In short, botulism causes flaccid paralysis; tetanus causes spastic paralysis.

Inflammatory infiltrate of T cells within muscle refers most notably to polymyositis and dermatomyositis.

43) A 51-year-old woman with a 15-year history of type II diabetes mellitus comes to the physician for a follow-up appointment. She has had gradually worsening vision over the past two years. HbA1c is 12%. A fundoscopy is shown. Which of the following is most likely to be seen in this patient?



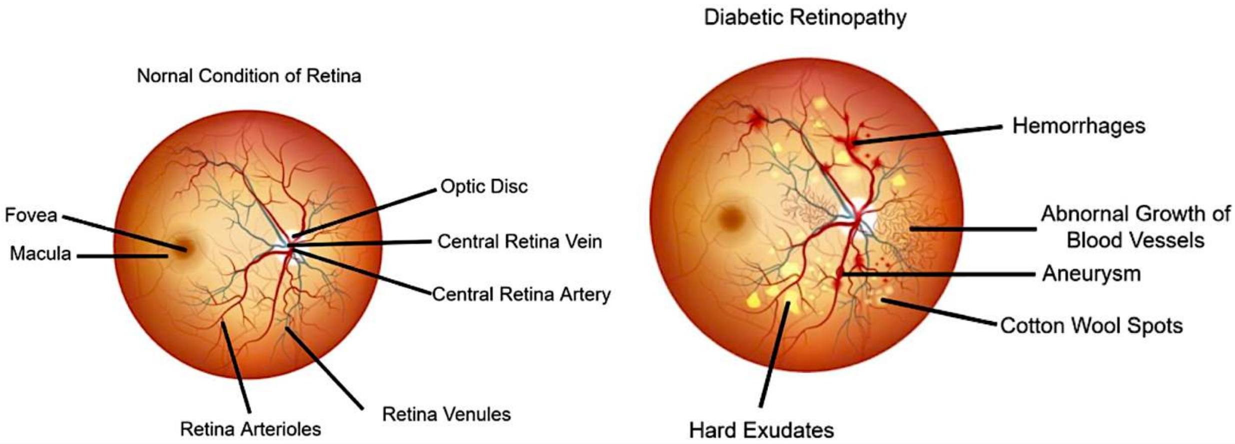
A) AV-nicking
B) Cotton wool spots
C) Cupping of the optic disc
D) Drusen
E) Papilledema

The answer is B (cotton wool spots).

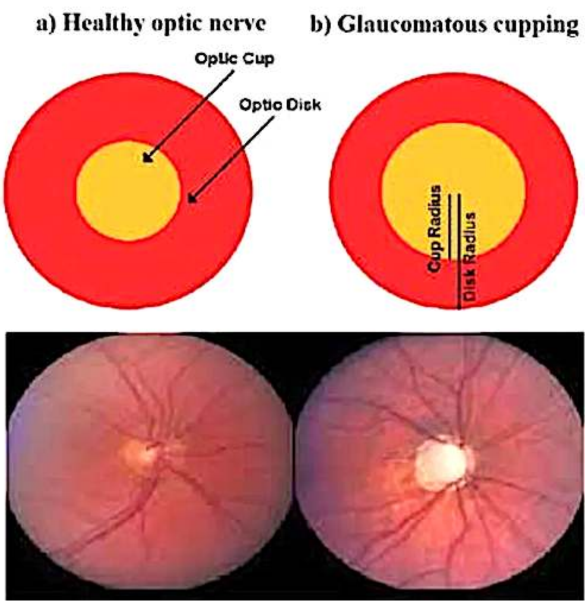
Cotton wool spots (soft exudates) are white, wispy, axoplasmic material due to microinfarcts within retinal vessels. They are seen in hypertension and diabetes.

AV (arteriovenous) nicking is a buzzy term that is a fundoscopic finding in hypertensive retinopathy.

Drusen is a yellow, lipoproteinaceous material seen as deposits on the retina in age-related macular degeneration. Don't confuse with hard exudates, which are lipoproteinaceous deposits seen on the retina in hypertension and diabetes.



Cupping of the optic disc is seen in glaucoma, which refers to an increase in the size of the cup compared to the optic disc. The optic disc is where the optic nerve exits the eye. The cup is a central depression within the optic disc with greater convergence of nerve fibers.



Papilledema is blurring of the optic disc margins seen with pseudotumor cerebri (increased intracranial pressure).

44) A 7-year-old boy is brought to the pediatrician for a follow-up appointment. He has history of recurrent fractures and joint hypermobility since birth. A Rinne test shows bone conduction greater than air conduction on the left. A Weber test lateralized to the left. Which of the following molecular mechanisms is most likely disrupted in this patient?

A) Incorporation of glycine into collagen type III primary structure
B) Hydroxylation of pro-collagen residues
C) Type IV collagen auto-antibodies
D) Type IV collagen gene mutation
E) Synthesis of glycoprotein that forms a sheath around elastin

The answer is B (Hydroxylation of pro-collagen residues).

Diagnosis is osteogenesis imperfecta (OI), which is a mutation in type I collagen. The USMLE doesn't care about the specific problem with collagen. There are many sub-types of OI with varying severity. Some are even fatal *in utero.* It is merely the case that choice B is the only answer that works.

Blue sclerae are hyper-pass-level and buzzy for USMLE. In fact, they are too easy and usually not a part of OI questions.

Kids usually present with fractures at different stages of healing because collagen I is the main constituent of organic bone matrix (i.e., outside of hydroxyapatite). Important differentials are child abuse and osteopetrosis.

Conduction hearing loss can occur due to abnormal formation of middle ear ossicles.

Conductive hearing loss on left: Rinne shows bone > air conduction; Weber lateralizes (is louder) to left.
Conductive hearing loss on right: Rinne shows bone > air conduction; Weber lateralizes to right.
Neurosensory hearing loss on left: Rinne shows air > bone conduction; Weber lateralizes to right.
Neurosensory hearing loss on right: Rinne shows air > bone conduction; Weber lateralizes to left.

In other words, air > bone conduction is both normal + also what we see in neurosensory hearing loss. Weber will lateralize to the ipsilateral ear in conductive hearing loss, and to the contralateral ear in neurosensory.

Incorporation of glycine into collagen type III (aka fibrillar collagen; don't confuse with fibrillin) primary structure refers to Ehlers-Danlos syndrome, which presents as a patient with hyperflexible joints and hyperextensible skin. There is increased risk of aortic dissection, mitral valve prolapse, and circle of Willis berry aneurysms.

Type IV collagen auto-antibodies (aka anti-GBM; anti-glomerular basement membrane) refers to Goodpasture syndrome, which will be a male 20s-40s with hematuria and hemoptysis. There is linear immunofluorescence on biopsy of the kidney.

Type IV collagen gene mutation is Alport syndrome. This presents as red urine in a male who has eye and/or ear problems. It is X-linked recessive.

Synthesis of glycoprotein that forms a sheath around elastin refers to fibrillin defect in Marfan syndrome. Autosomal dominant; *FBN1/2* mutation on chromosome 15.

45) A 29-year-old man with a history of schizophrenia is commenced on a new medication. Several days after initiation, he presents to the physician with acute episodes of sustained, involuntary upward eye deviation lasting minutes to hours. During these episodes, he experiences severe neck and trunk muscle spasms as well. Vitals are normal. Neurological examination is otherwise unremarkable. Which of the following is the most likely ophthalmologic diagnosis?

A) Blepharospasm
B) Endophthalmitis
C) Ectropion
D) Entropion
E) Oculogyric crisis
F) Opsoclonus-myoclonus syndrome

The answer is E (oculogyric crisis).

The patient has acute dystonia from commencing an anti-psychotic medication. This can present as abnormal eye movements (oculogyric crisis), torticollis (stiff, crooked neck), and muscle rigidity (without fever; if patient has fever >102-3F, think neuroleptic malignant syndrome). Treatment is with an anti-cholinergic medication such as benztropine (direct muscarinic receptor antagonist) or diphenhydramine (1^{st}-gen H1-blocker with nasty anti-cholinergic side-effects, but the latter are “good” when we want to treat acute dystonia).

Blepharospasm is involuntary and repetitive blinking or eyelid spasm; can be associated with difficulty with eyelid closure.

Endophthalmitis is a deep infection of the eye, usually due to penetrating trauma or as a post-op sequela.

Ectropion is when the lower eyelid turns outward away from the eye.

Entropion is when the lower eyelid rolls inward toward the eye, causing the eyelashes to rub against the eye's surface.

Opsoclonus-myoclonus syndrome is rapid, chaotic eye movements (opsoclonus) and sudden, involuntary muscle jerks (myoclonus). On USMLE 2CK pediatrics, it is seen in neuroblastoma.

46) A 28-year-old woman presents to the physician after a night of heavy partying. Her left eye is shown. Which of the following is the most likely diagnosis?

A) Conjunctivitis
B) Corneal abrasion
C) Hyphema
D) Scleritis
E) Subconjunctival hemorrhage

The answer is E (Subconjunctival hemorrhage).

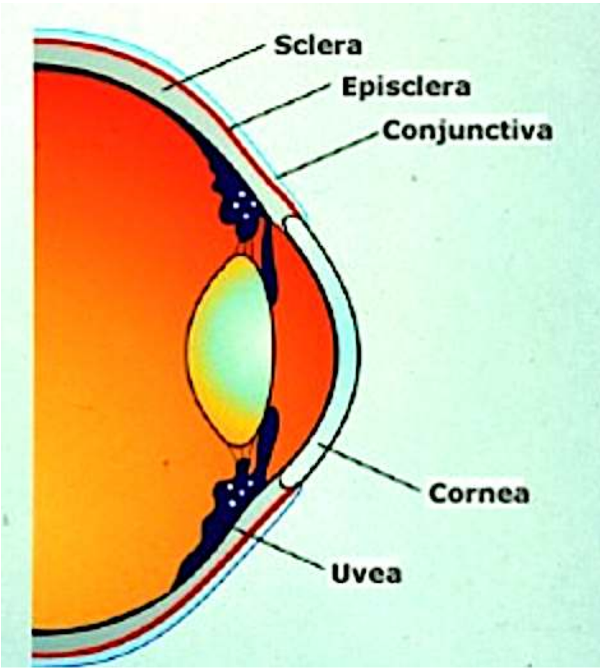
Increased pressure in the eye from vomiting presumably caused breakage of superficial conjunctival vessels leading to a bleed. This is pass-level spot-diagnosis, but nevertheless needs to be covered.

Conjunctivitis has many types on USMLE. Chemical conjunctivitis is due to silver nitrate eye drops that used to be used in neonates. Chlamydial conjunctivitis tends to occur after the first week of life and leads to pneumonia. Gonococcal conjunctivitis presents usually in the first week of life and doesn't cause pneumonia. Allergic conjunctivitis presents with "cobblestoning" of the tarasal conjunctiva. Viral conjunctivitis is due to adenovirus. Bacterial can be polymicrobial or *S. aureus*.

Corneal abrasion will be shown on USMLE as a green area on a blue eye with fluorescein instillation.

Hyphema is blood in the anterior chamber.

Scleritis is inflammation of the white tough coating of the eye composed of dense, fibrous tissue. This can be seen classically in autoimmune disease, same as for anterior uveitis or episcleritis.



The conjunctiva is a thin, transparent mucous membrane that lines the inner surface of the eyelids (tarsal or palpebral conjunctiva) and extends to cover the front surface of the sclera (bulbar conjunctiva), providing protection and lubrication for the eye.

The episclera is a thin, transparent layer of tissue that lies just beneath the conjunctiva and covers the sclera. It contains blood vessels and helps nourish the sclera.

The anterior uvea is the combination of the ciliary body (muscle that controls the lens) and the iris.

47) A 39-year-old man comes to the physician for a 1-week history of yellow eyes. He had an appendectomy performed two weeks ago. Laboratory studies show: direct bilirubin 0.1 mg/dL (normal ~0.1); total bilirubin 3 mg/dL (normal ~1.0). Serum LDH is normal. Which of the following is the most likely explanation for these findings?

A) Gilbert syndrome
B) Hemolysis
C) Dubin-Johnson syndrome
D) Ocular albinism
E) Rotor syndrome

The answer is A (Gilbert syndrome).

Gilbert syndrome is benign unconjugated hyperbilirubinemia that occurs due to deficiency of UDP-glucuronosyltransferase, the uptake enzyme for bilirubin at the liver. This can lead to recurrent episodes throughout life following stress, e.g., while studying for exams or post-surgery. Normal serum LDH is an overt negative for hemolysis, since high LDH is very buzzy for hemolysis if a Q says it (RBCs are packed with LDH).

Dubin-Johnson and Rotor syndromes are conjugated hyperbilirubinemias due to defective secretion of bilirubin into bile. The liver can appear black in DJS and normal in Rotor syndrome.

Ocular albinism clearly makes no sense and would present as pale blue or pink irides (fancy word that is pleural for iris) in neonates.

The sometimes-pink appearance of the eyes is due to visibility of blood vessels due to lack of pigmentation in the iris and the translucency of the ocular structures.

48) A 28-year-old man comes to the physician in August for a 1-week history of unsteady gait and visual disturbance. He is an avid traveler and spent most of the year driving between countries in Europe. Physical examination shows difficulty standing with his eyes closed. Ophthalmologic examination shows pupils that change size with moving objects near the eye but do not change size in response to light. Which of the following is the most likely explanation for these ophthalmologic findings?

A) Argyll-Robertson pupils
B) Internuclear ophthalmoplegia
C) Marcus-Gunn pupils
D) Tabes dorsalis
E) Weber syndrome

The answer is A (Argyll-Robertson pupils).

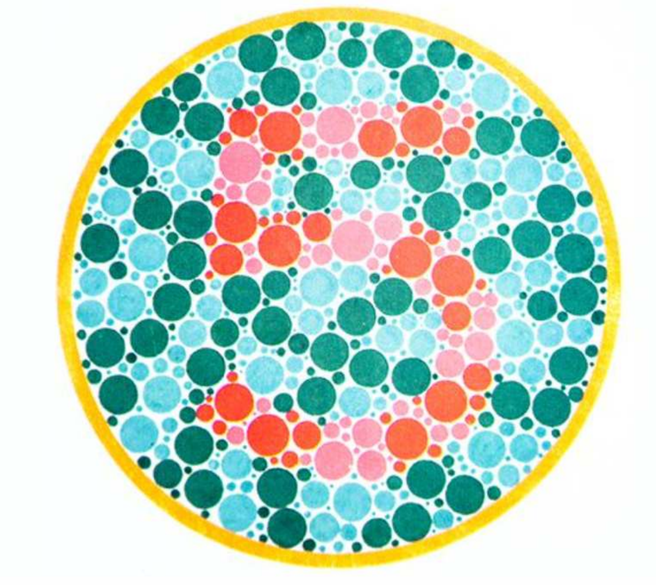
Diagnosis is neurosyphilis. Argyll-Robertson pupils are colloquially known as “prostitute’s pupils” because they accommodate (to object distance) but don’t react (to light).

The patient has a positive Romberg sign, which indicates the etiology for his ataxia is the dorsal columns. This is due to tabes dorsalis as part of the neurosyphilis. But the tabes dorsalis itself is not the cause of the eye findings. The question asks for the most likely explanation for these ophthalmologic findings.

Internuclear ophthalmoplegia and medial longitudinal fasciculus syndrome are the same thing, which is how you know they’re both fucking wrong. This is seen in multiple sclerosis on USMLE. I talked about INO/MLF syndrome in detail earlier in this PDF.

Weber syndrome is a midbrain stroke that causes ipsilateral CN III palsy and contralateral spastic paresis of the body. It shows up on the NBME exam for Step 1. But “showing up” doesn’t = HY. There’s like one question on it out of thousands.

49) A 28-year-old man from Indonesia undergoes a physical prior to enrollment in the military. He is a routine heavy partier and uses various drugs. He cannot read the number within the yellow circle below. Which of the following is the most likely cause of these findings?



A) Digoxin
B) Ethambutol
C) Heroin
D) Lysergic acid diethylamide (LSD)
E) Phencyclidine (PCP)
F) Rifampin
G) Sildenafil

The answer is B (ethambutol).

Ethambutol is a drug used to treat tuberculosis that can cause red-green color-blindness.

Digoxin can cause yellow, wavy “Vincent van Gogh vision.”

Heroin and opioids cause pinpoint pupils. Organophosphates (unrelated) are also a HY cause of pinpoint pupils on USMLE.

LSD can cause visual hallucinations.

PCP causes vertical and horizontal nystagmus.

Rifampin is a drug used to treat tuberculosis that causes red-orange tears and sweat.

Sildenafil (Viagra) can cause blue haze. Pilots must avoid taking this prior to flying.

50) A 44-year-old woman comes to the physician because of a rash on her hands. Which of the following additional findings is most likely to be seen in this patient?

A) Abnormal blinking
B) Orbital xanthomata
C) Pronounced limbal rings
D) Robust corneal arcus
E) Violaceous eyelids

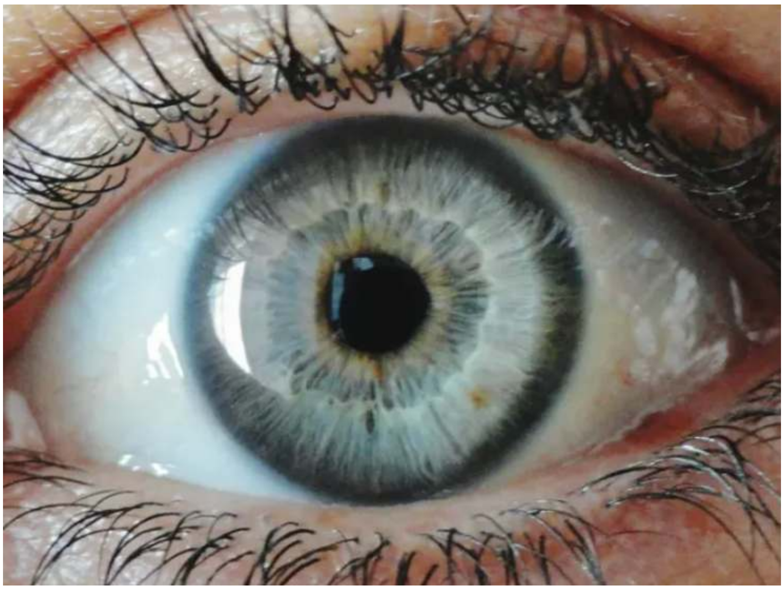
The answer is E (violaceous eyelids).

The image is a spot-diagnosis for Gottron papules in dermatomyositis. Patients can also get heliotrope rash, which refers to violaceous eyelids or a periorbital rash. **Do not confuse this with the malar rash in SLE.**

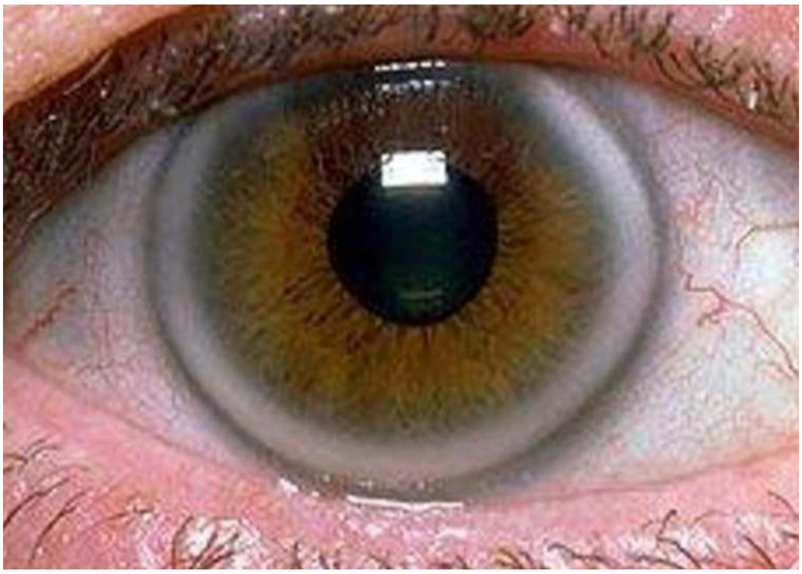
Abnormal blinking could be seen in absence seizure, Tourette syndrome, or as a provisional tic disorder in pediatrics.

Orbital xanthomata (xanthelasma) are lipid deposits seen periorbitally in patients either idiopathically or secondary to dyslipidemia.

Limbal rings refer to the distribution of greater density of pigmentation circumferentially within the iris. These are a normal finding and are more pronounced in the youth.



Corneal arcus is lipid deposition seen circumferentially within the iris in patients either idiopathically due to age or secondary to dyslipidemia.



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